LASERCHECK

User Guide

Everything LaserCheck can do, and how to do it: editing, markup, measuring drawings, forms, signing, redaction, building document packs and checking reports before they go out.

Version 1.34.0

Mac · Windows · Browser

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1 Welcome

LaserCheck is a PDF editor built for people who work with drawings, field reports and handover documents. It runs on your own computer, so your documents never leave it.

Edit anything

Change existing text and whole paragraphs, add text in 35 bundled font families, insert pictures, cover things up, and add links.

Mark up and measure

Highlight, draw, cloud and stamp drawings. Set a scale and measure lengths, routes and areas, and count items.

Forms and signing

Fill in and create forms, fill them from your saved details, mail-merge from a spreadsheet, sign, and seal documents with a trusted timestamp.

Packs and checks

Build a handover pack from a folder in one step, and check a field report the way a fussy client would before you send it.

The desktop app and LaserCheck Lite

LaserCheck is a desktop app for Mac and Windows. Everything in this guide is about the desktop app. **LaserCheck Lite** is a free version that runs in a web browser: it opens, views, searches and prints PDFs and shows this guide, but doesn't edit or save.

FEATURE	DESKTOP APP	LASERCHECK LITE
Open, view, search and print PDFs; this guide	**Yes**	**Yes**
Editing, markup, measuring, forms, signing, pages, packs, the report checker	**Yes**	No
Saving and exporting	**Yes**	No

The desktop app is free to use for now, with everything switched on.

Welcome, what's new and the sample documents

The first time LaserCheck opens it shows the main things it does, how to do each, and a **Try it** button that opens a sample and shows the tool. **Help > Welcome and Main Features** shows it again.

After an update, the first launch lists **what's new** since the version you used before. **Help > What's New** shows the recent changes at any time.

Help > Samples… offers sample documents that each show off one tool:

SAMPLE	TRY
HV and fibre report	Check Reports at the top: an HV report with fibre results bound in, typed on a phone. Both parts are checked in one go, typos included.
Field report with mistakes	Check Reports at the top: a wrong serial in a photo, a reused photo, an expired calibration, results under the pass criterion and more.
Floor plan with layers	The **Layers** button at the bottom: show the hidden layer.
German procedure	**Translate** (in Pages): an English copy with the same layout.
Inspection form (not fillable)	**Detect fields** (in Forms): make it fillable.
Altered certificate	**History and authenticity** (in Pages): see the word that was changed after it was made.

Samples open as copies, so change them freely.

Unsaved work after a crash

While a document has edits that aren't saved, LaserCheck keeps a copy of it on this computer about once a minute. If it closes in any other way than quitting (a crash, a power cut), the next start offers to open the copies, each with "(recovered)" in its name: save it to keep it. Saving, closing a document or quitting normally throws the copy away. A recovered document has your edits built into its pages; added text can still be changed with Edit text.

Updates

LaserCheck checks for a new version each time it starts (and every few hours while it's open) and downloads it in the background. When it's ready, a card offers **Install and restart**: LaserCheck closes (asking about any unsaved documents first), the update installs, and LaserCheck opens again. On a Mac, the

installer asks for your password as usual. Choose **Later** and it installs the next time you choose Install, from **Check for Updates…** (in the **LaserCheck** menu on a Mac, the **Help** menu on Windows).

Every update is signed by LaserCheck, and its download is checked against the signature before it's used, so a damaged or tampered download is thrown away. Turn automatic checks off in the same place.

2 Installing

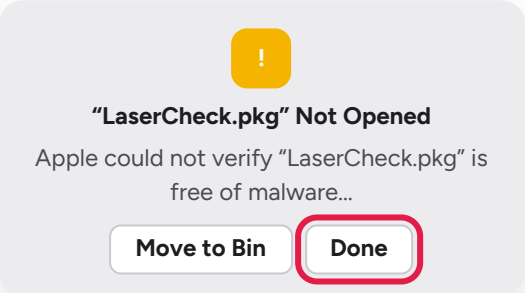
Install once, and LaserCheck works offline from then on.

Opening it the first time

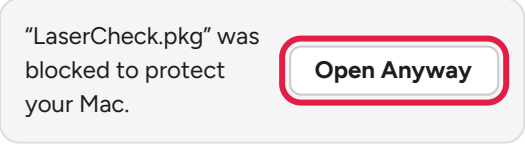
Mac and Windows both ask once before running a newly downloaded app. This is what you'll see, and what to click (the pictures are simplified).

On a Mac

1. **Open the downloaded file.** The first time, macOS says it can't check it. Click **Done**.



2. **Open System Settings › Privacy & Security**, scroll down, and click **Open Anyway**, then enter your password.

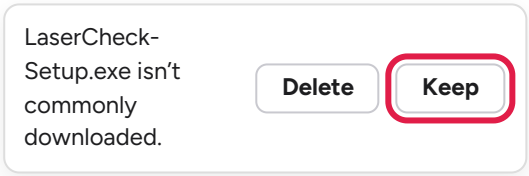


3. **Install as usual.** LaserCheck goes in Applications. You only do this once: updates install without asking.

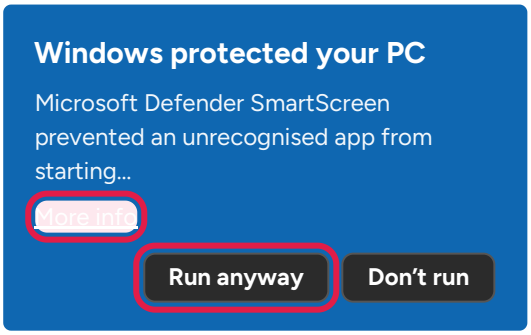
On older macOS, you can instead right-click the file and choose **Open**. Updating from Laser PDF? Open **LaserCheck** from Applications once afterwards.

On Windows

1. **If your browser says the file isn't commonly downloaded,** choose **Keep** (in Edge: the ⋯ menu, then **Keep**, then **Keep anyway**).



2. **Run the file.** If a blue box says **Windows protected your PC**, click **More info**, then **Run anyway**.



3. **Setup finishes on its own**, with no administrator rights needed. You only do this once: updates install without asking.

These warnings appear because LaserCheck isn't yet signed with a paid Apple or Microsoft certificate. It's the same app either way.

Mac

1. Double-click **LaserCheck.pkg**.
2. If your Mac says it can't check the app for malicious software, open **System Settings → Privacy & Security**, scroll down, and click **Open Anyway**. You only do this once. (Or, in Terminal: `xattr -d com.apple.quarantine ~/Downloads/"LaserCheck.pkg"`.)
3. Follow the installer. It downloads the app's runtime (about 130 MB) from the Electron project, checks it against its published checksum, then builds the app on your Mac and puts it in **Applications**. You need an internet connection for this step only.

If the package won't run, **Install LaserCheck.command** does the same job: right-click it, choose **Open**, and follow the prompts.

> **TIP**
> To open PDFs in LaserCheck by default: select any PDF in Finder, press ⌘ I, choose LaserCheck under **Open with**, then click **Change All…**.

Windows

1. Double-click **LaserCheck Setup.exe**.
2. If Windows SmartScreen appears, click **More info**, then **Run anyway**.
3. The installer puts LaserCheck in your Start menu. It installs for you only, so it doesn't need an administrator.

To remove it, use **Settings → Apps → Installed apps → LaserCheck → Uninstall**.

LaserCheck Lite

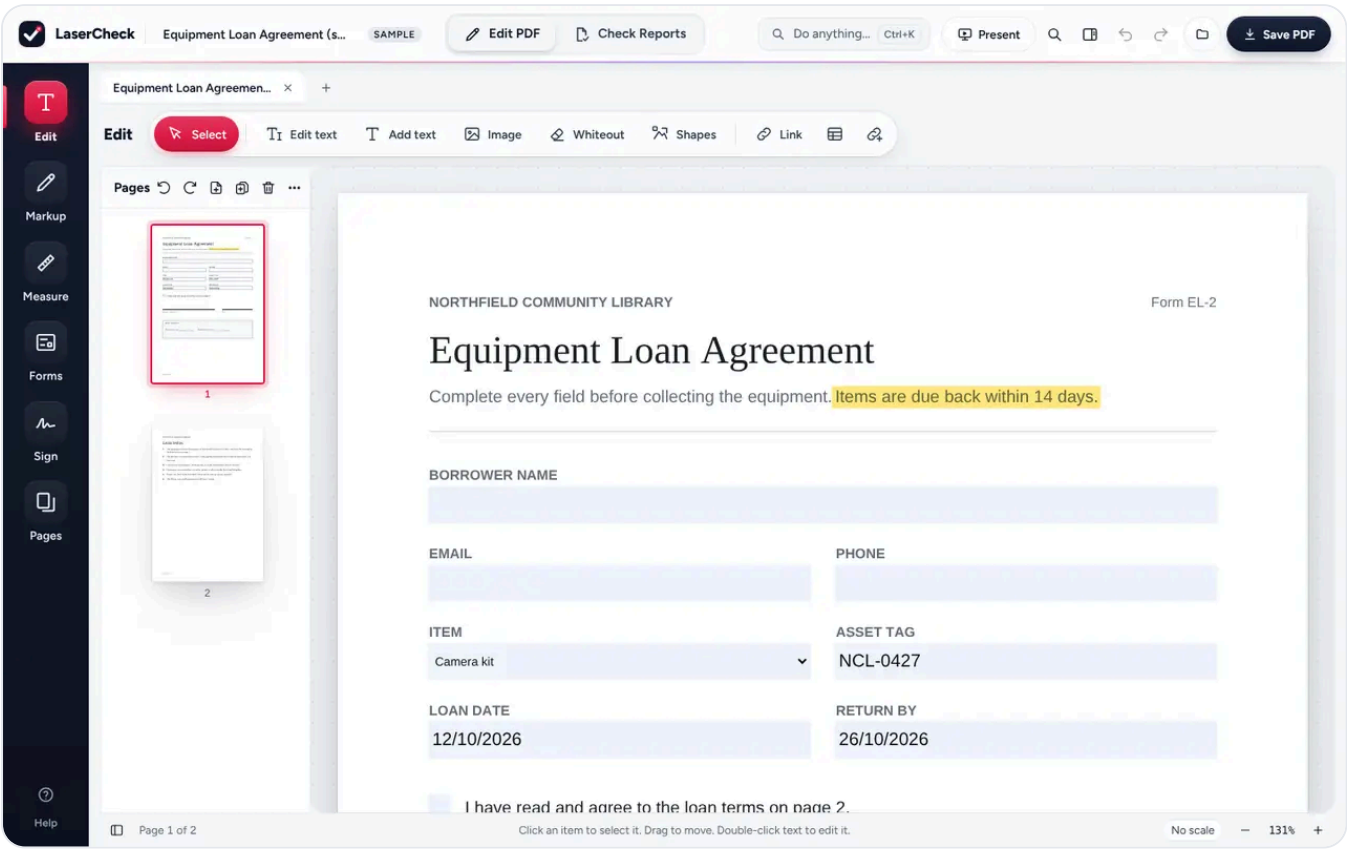
Open **LaserCheck Lite.html** in Chrome, Edge, Safari or Firefox. Nothing is installed and nothing is uploaded. It's handy for looking at PDFs on a computer where the desktop app isn't installed.

Updating

Run the new installer over the old one. Your licence, settings, saved signature, templates and recent files are kept. Every licence includes all updates.

3 The workspace

One window, two workspaces: **Edit PDF** and **Check Reports**, switched at the top of the window. Editing has six modes, each with its own toolbar, so you only see the tools for the job in hand.



A document open in Edit mode: the modes down the left, that mode's tools above the page, page thumbnails beside it.

Down the left

- **Modes** (in Edit PDF): Edit, Markup, Measure, Forms, Sign and Pages. Switch with ⌘ 1 to ⌘ 6.
- **The checker's tools** (in Check Reports): HV or Fibre at the top, then the tools for that kind of report, grouped into This report, Check against and Whole job. See the report checker.
- **Ask AI** (desktop app): questions about the document.
- **Help**: this guide.

Across the top

- **Document name**: click it to rename the file you'll save.
- **Edit PDF | Check Reports**: the two workspaces. Once a report has been checked, Check Reports shows how many things need looking at.
- **Do anything...** (⌘ K): type what you want to do. See Do anything.

- **Find** (magnifier), the **side panel** button, **Undo** and **Redo**.
- **Open** and **Save PDF**.

The status bar along the bottom shows the page, the zoom and a hint for the tool you're using.

Do anything (⌘K)

Press ⌘ K (or click **Do anything…** at the top) and type a few letters of what you want: “rotate”, “word”, “redact personal”, “scale”. Initials work too (“cpr” for Compare procedure revisions). Use ↑ ↓ and Enter, or click. Every command, mode and tool is there, with its shortcut, and the ones you use most come first. You can also type:

- **page 12** to go to a page, **zoom 150** to zoom;
- **find** followed by words to search for them;
- **redact** followed by words to mark them for redaction everywhere.

The modes

Edit
Change and add text, pictures, whiteout, links, tables to Excel.

Markup
Highlight, draw, shapes, clouds, comments, stamps, QR codes.

Measure
Scale, distance, path, area and count, with a quantities list.

Forms
Create fields, fill in your details, mail merge.

Sign
Signature, seal, redaction and password protection.

Pages
Organise pages, headers and numbers, combine, compare, export, packs, checks.

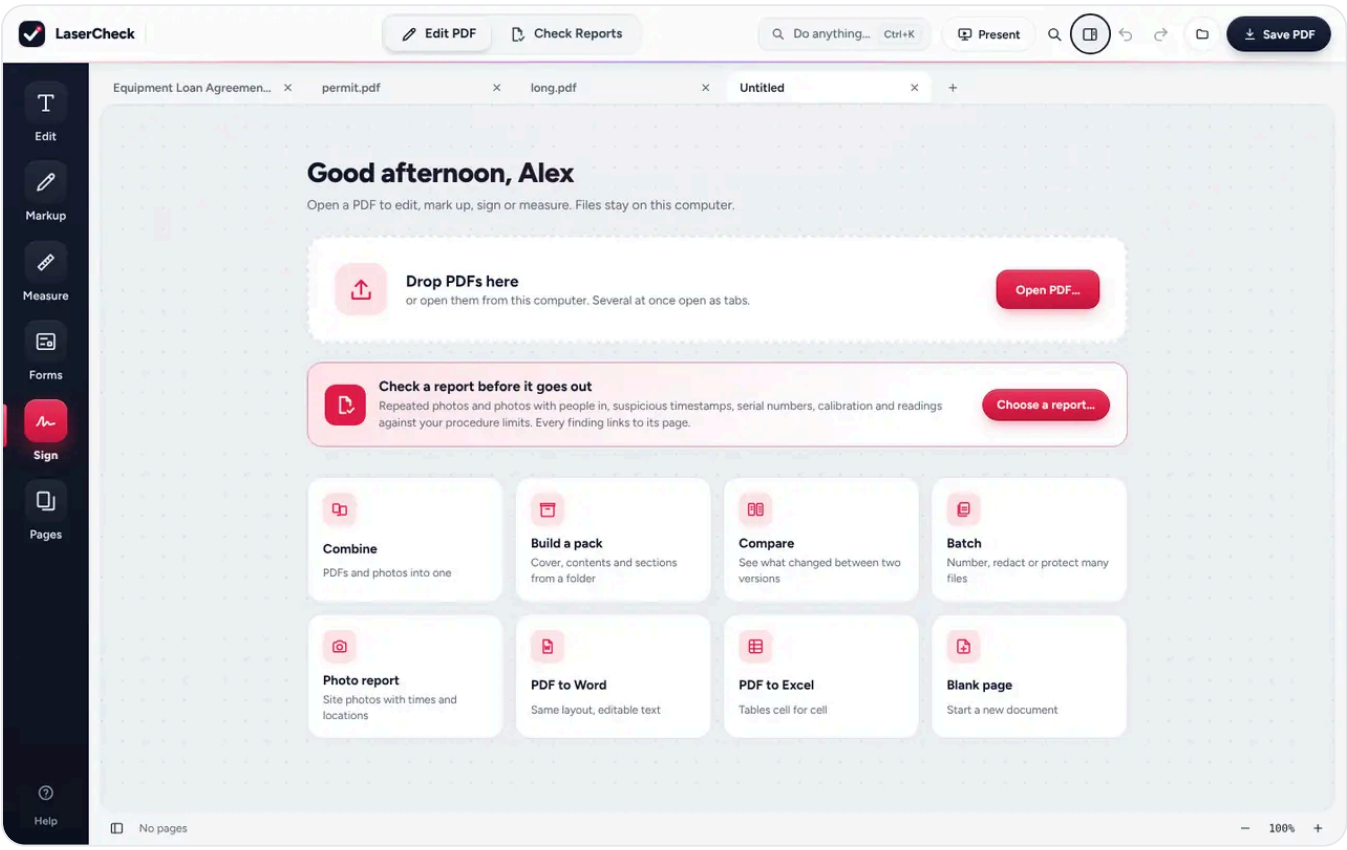
Panels

- **Pages** (left): thumbnails of every page. Click to jump; drag to reorder in Pages mode. Show or hide with ⌥ ⌘ P.
- **Side panel** (right): **Comments**, **Quantities**, **Bookmarks** and **Checks**, as tabs.
- **Status bar** (bottom): the current page, the drawing scale, and zoom.

Tabs

Each document opens in its own tab, so you can work on several at once. ⌘ T opens a new tab, ⌘ W closes one, and Ctrl Tab moves between them.

The start screen



The start screen: open or drop files, quick actions, and your recent files.

A new tab shows the start screen, with one-click **Combine**, **Build a pack**, **Compare**, **Batch**, **Photo report** and **Blank page**, and lists the files you opened or saved recently. Hover over one and click × to take it off the list.

Light and dark

LaserCheck follows your computer's light or dark appearance automatically.

4 Opening and saving

Opening

- Click **Open** or press ⌘ O, or drag PDFs onto the window.
- You can also double-click a PDF in Finder or File Explorer, or pick from **Recent files** on the start screen.
- To turn pictures (JPEG, PNG and others) into PDF pages, use Combine.

If a PDF is password-protected, LaserCheck asks for the password. If it's digitally signed, it tells you so: editing a signed PDF breaks its signature, so it's best to work on a copy.

Saving

- **Save** (⌘ S) writes your changes back into the same PDF.
- **Save As** (⇧ ⌘ S) saves under a new name.
- **Save a Smaller Copy** (File menu, or **Reduce file size** in Pages mode) shrinks big scans and photos. See Pages and documents.

Comments, markup and form entries are saved as standard PDF annotations, so they show up in Adobe Acrobat, Preview, Bluebeam and other readers. If you close a tab with unsaved changes, LaserCheck asks first.

Timeline: earlier versions

File → Timeline… (⌥ ⌘ T, or click **Saved** or **Edited** in the status bar) shows the document's history.

- **Since you opened it**: every edit, on a slider. Drag it to see the document as it was at that point. **Go back to here** keeps it, and Redo still brings the later edits back. Close without choosing and nothing changes.
- **Saved versions**: the desktop app keeps a copy each time you save, and the file as it was before your first save over it, so the original is never lost. **Open** opens one as a separate copy, **Compare** shows what changed against the document now, and **Restore** puts it back (save to keep it; the version you replace stays in the timeline).

Versions are kept on this computer only: up to 50 for each file and 2 GB in all, the oldest going first.

Printing

Press ⌘ P, choose **File → Print…**, or click **Print** on the Pages toolbar. Choose all pages, the current page, or a list such as `1-3, 6`, and the quality: **Standard** for everyday documents or **High** for drawings. LaserCheck prints the document as it looks now, with your changes, comments and markup, and each page prints at its own size and way round. Then pick the printer and paper in your computer's print window, where you can also save as PDF. Printing works during the trial, after it, and in LaserCheck Lite.

UNDO

Almost everything can be undone with ⌘ Z and redone with ⇧ ⌘ Z. Each tab keeps its own history.

5 Editing text and content

Edit mode (⌘ 1) changes what's actually on the page.

Editing existing text E

1. Choose **Edit text**, then click the text you want to change.
2. LaserCheck picks up the paragraph you clicked (or just the word or line: see below), and its font, size and colour as closely as it can. Type your changes; the paragraph re-wraps as you go.
3. Click away, or press **Esc**, to finish.

A small bar above the box lets you choose what to edit: **Word** (the word you clicked is selected, so just type its replacement), **Line** or **Paragraph**. LaserCheck remembers your choice for the next click. Holding ⌥ (Alt on Windows) as you click always edits just the line. To change part of the text, **drag across it** instead of clicking: the words you sweep over are highlighted, and when you let go just those lines open with exactly that text selected, so you can type straight over it. When you save, the original words are removed from the file, not just covered up, so they can't be copied or found by search.

GOOD TO KNOW

- A line that mixed fonts (say bold and normal) comes back in one font.
- Justified paragraphs become left-aligned. Centred titles stay centred.

Adding text T

Choose **Add text**, click where it should go, and type. Use the bar that appears to pick the font, size, colour, bold and italic. LaserCheck includes 35 font families (sans, serif, condensed, monospace and handwriting styles); it also lists the fonts installed on your computer. Fonts are embedded in the saved PDF, so it looks the same everywhere.

Typing into a table

Many blank reports are tables drawn with lines rather than forms. Click an empty cell with **Edit text** or **Add text** and the new text starts lined up in that cell: in the font, size and colour of the text already in the column (or row), the same distance in from the cell's left line, and on the row's baseline. Clicking a cell that has text in it edits that text as usual, and clicks away from a table work as before.

Copy formatting

Press F (or **Copy formatting** on the text bar, or **Edit → Copy Formatting**), click the text whose look you want, then click each text that should match it: its font, size, bold or italic, and colour are copied. With a text box selected first, its look is copied straight away. Page text you click becomes editable text in that look, the same as Edit text would make it. It stays on until you press Esc; each change is one Undo.

Lining text up

Faint dashed guide lines show when text lines up with the text around it: the start and end of the page's own lines, their baselines, the middle of a line, other text boxes, and the centre of the page. A new text box starts on the nearest line start and baseline. When you drag a text box near a guide it snaps onto it; hold ⌥ (Alt on Windows) while dragging to move it freely. While you type, the guides show what the box already lines up with.

When you edit the page's own text, it's retyped in the same font where LaserCheck has it, or the closest-looking one where it doesn't (Verdana, Segoe UI, Century Gothic, Garamond, Consolas and many others are matched to a similar face).

Pictures I, whiteout W and links

- **Image**: place a picture or logo, then drag its corners to size it.
- **Whiteout**: drag a box to cover something with white (or another colour). To *remove* sensitive content properly, use redaction instead.
- **Link**: drag over words or an area, then choose a web address, email or a page in the document.
- **Auto-link addresses**: makes every web and email address in the document clickable in one go.

Layers

Drawings exported from CAD often come in layers. When a document has them, a **Layers** button appears at the bottom of the window (also **View → Layers…**). Switch layers on and off; what you see is what prints and exports, and saving keeps it as the file's view. To take hidden layers out of the file for good, use Flatten.

Shapes: edit the page's own drawing

Shapes works on the lines, boxes, shading and pictures that are part of the PDF itself, not markup added here. Click one to select it; drag round several, or Shift-click to add more.

- **Move**: drag, or nudge with the arrow keys (hold Shift for 10 pt).
- **Restyle**: change the line colour, width or style (solid, dashed, dotted) and the fill colour from the bar above the page.
- **Delete**: press ⌫, or click **Delete**.

- **Select similar** picks every shape on the page drawn the same way, so you can, say, turn all the red lines on a drawing black in one go.

Text isn't touched (use **Edit text** for that), and undo works as usual. Pages with form fields, and PDFs that limit changes, can't be edited this way. Some PDFs group a whole drawing into one object: then it moves or deletes as one.

Find and replace

⌘ F searches the whole document; ⌘ G and ⇧ ⌘ G step through the matches. To change a detail everywhere (a job number, client name or date), open **Find & replace** with ⌥ ⌘ F. It shows how many matches there are as you type, with **Match case** and **Whole words** options, then **Replace all** updates every page.

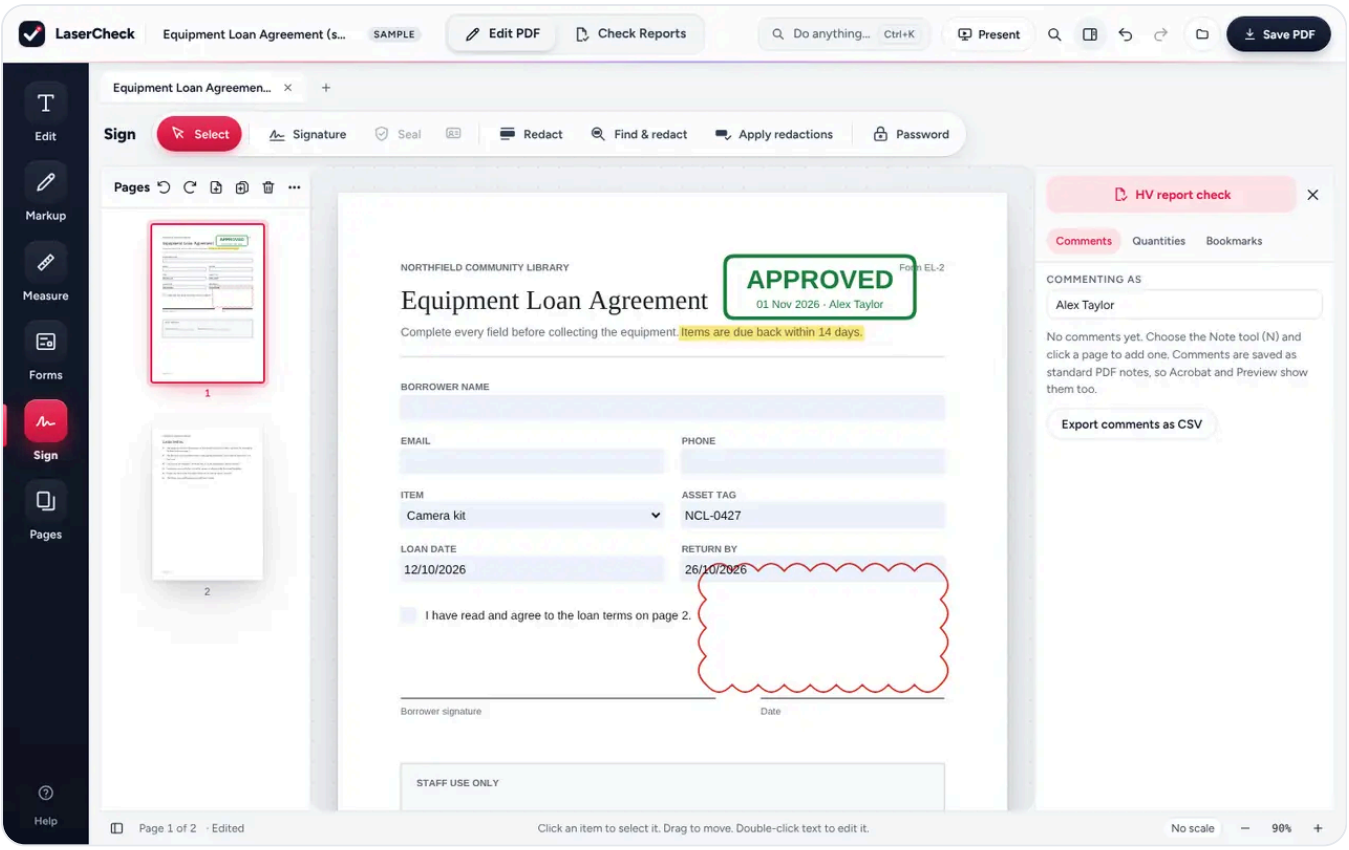
> GOOD TO KNOW
>
> Find & replace matches text within a line, so a phrase that wraps onto the next line won't be found. Scanned pages need text recognition first.

Table to Excel

Choose the table tool (in Edit mode, or **Tools → Table to Excel**), then drag a box around a table, or click to take the whole page. A preview shows the rows and columns it found. Tick **Join wrapped lines** if cells run over two lines, and **First row is headings** if they are. Then **Copy** (to paste into a spreadsheet), **Save CSV** or **Save Excel**. Numbers, thousands separators and percentages come through as real numbers.

6 Markup and comments

Markup mode (⌘ 2) is for reviewing: everything here can be moved, restyled and removed later.



A highlight, a revision cloud, a stamp and a comment.

TOOL	KEY	HOW TO USE IT
Highlight	H	Drag across text.
Draw	D	Draw freehand.
Rectangle, Ellipse, Line, Arrow	R O L A	Drag to draw. Hold ⇧ for squares, circles and straight lines.
Revision cloud	C	Drag a box to cloud an area, the usual way to flag a change on a drawing.
Comment	N	Click to pin a note and type. Others can reply.
Stamp	B	Click to place: APPROVED, AS BUILT, CHECKED, CONFIDENTIAL, DRAFT, FOR CONSTRUCTION, FOR INFORMATION, FOR REVIEW, NOT APPROVED, REDLINE, SUPERSEDED, VOID, or **Custom...** for your own wording.
QR code		Click or drag to place a QR code for a web link, any text, or a map location, with an optional caption underneath.

More tools

The **More tools** button, next to the shapes, opens these:

TOOL	KEY	HOW TO USE IT
Underline, Strikethrough	U X	Drag across words to underline them or strike them through, for reviewing wording.
Callout	Q	Drag a box, type the note in the bar above, then drag the round handle to point the arrow at what you mean.
Text box		A bordered box of text, without an arrow.
Polygon, Polyline	Y J	Click each corner; press **Enter** or click the last point again to finish. A polygon closes and can be filled; a polyline stays open (a route or run). Drag a corner afterwards to adjust it.
Double arrow		A line with arrowheads at both ends, for dimensions and extents.
Revision triangle		Click to place the lettered Δ used on drawing revisions, then type its letter or number.
Tick, Cross		Click to tick something off, or cross it out, when checking a drawing or schedule.

Rectangles, ellipses, lines, arrows, polygons and polylines can be **solid, dashed or dotted**, and lines and arrows can have arrowheads at **the end or both ends**, using the bar above the page.

Select any item to change its colour, line width, opacity or text in the bar above the page. Drag to move it, drag its handles to resize, use the arrow keys to nudge (⇧ for bigger steps), ⌘ D to duplicate, and ⌫ to delete.

The Comments panel

Open it with the side panel button or ⌥ ⌘ C. It lists every comment with its author and date. Click one to jump to it, or reply in place. **File → Export Comments as CSV** gives you a list for a spreadsheet or a comments log.

> **TIP**
> The first time you comment or sign, LaserCheck asks for your name. It's shown on your comments and kept on this computer only.

Reading QR codes

Read QR codes (Markup toolbar, or **Tools → Read QR Codes**) finds every QR code in the document. For each one you can **Show** it on the page, **Open** or **Copy** it, and **Make links** turns them all into clickable links.

7 Measuring drawings

Measure mode (⌘ 3) takes off lengths, routes, areas and counts from drawings at their real scale.

1. Set the scale

- **Measure → Set Drawing Scale…** (or the scale button on the toolbar): enter the drawing's scale, such as 1 : 100, and choose whether it applies to **This page**, **All pages of this size** or **All pages**.
- If you don't know the scale, or the PDF was printed to fit: **Calibrate from a known length**, drag along a dimension you know (a grid line, a dimensioned wall), and type its real length.

The status bar shows the scale of the current page.

2. Measure

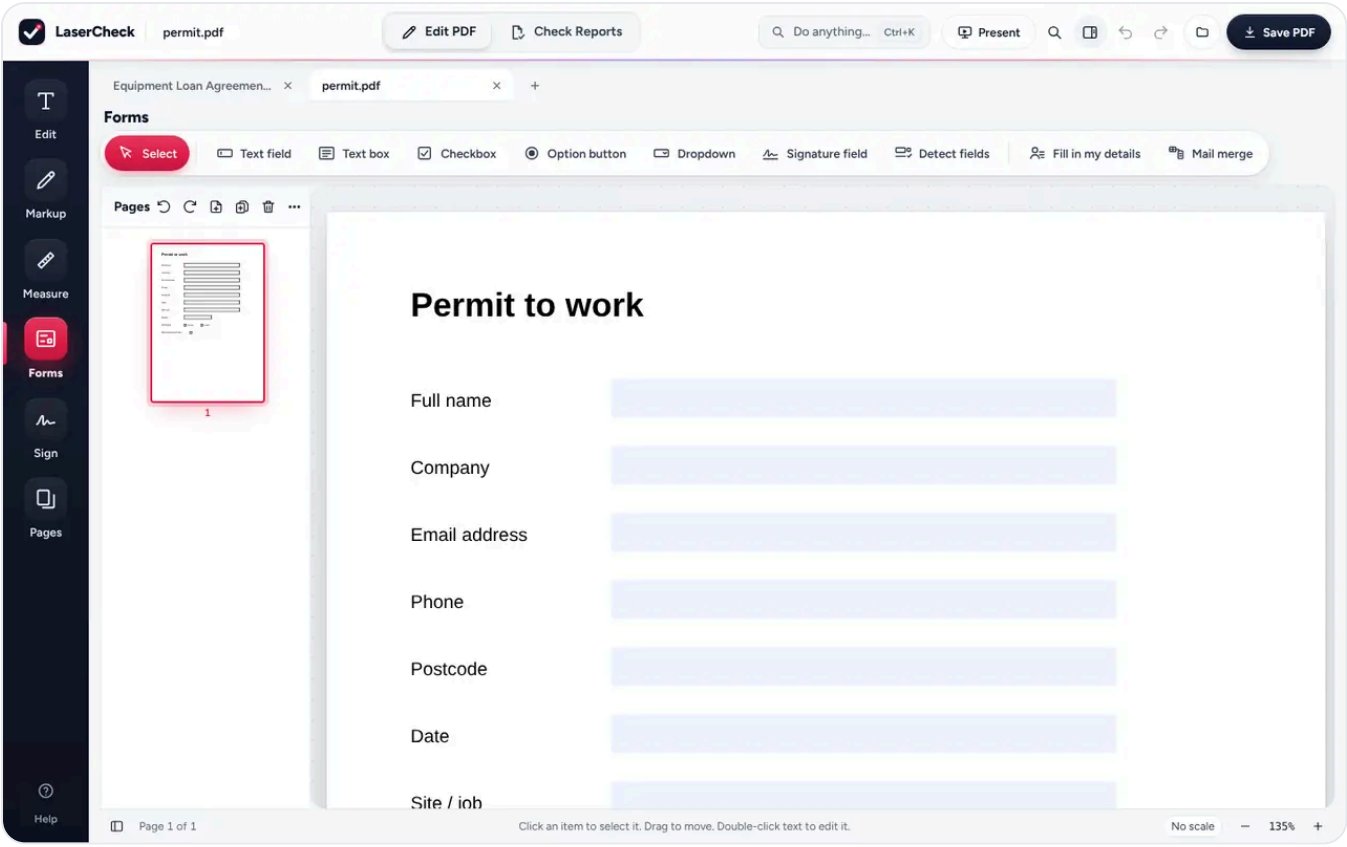
TOOL	KEY	HOW TO USE IT
Distance	M	Drag from one point to another.
Path	P	Click each point along a route (a cable run, a duct); press Enter or double-click to finish. ⌫ removes the last point.
Area	G	Click round the outline; press Enter or double-click to close it.
Count	K	Click each item (chambers, poles, sockets) to count it. ⌥-click a marker to remove it.

Drag any point afterwards to adjust a measurement.

3. Quantities

The **Quantities** panel (⌥ ⌘ Q) totals everything you've measured and counted. **File → Export Quantities as CSV** gives you the take-off for a spreadsheet.

8 Forms



Forms mode: field tools on the toolbar, Fill in my details and Mail merge on the right.

Filling in a form

If a PDF has form fields, just click them and type, in any mode. Tick boxes and choose from lists the same way. Your entries are saved into the form when you save.

Making a form

In Forms mode (⌘ 4), choose a field type and drag it onto the page: **Text field**, **Text box** (several lines), **Checkbox**, **Option button**, **Dropdown** or **Signature field**. Select a field to name it and set its options. The result is a standard fillable PDF.

Detect fields

Got a form that was made to be printed? Click **Detect fields** in Forms mode. LaserCheck finds the empty boxes, the empty table cells beside (or under) a label, the lines to write on and the tick boxes, and turns them into fillable fields named after their labels: “Inspector”, “Cable number”, “Signature” and so on. Check them over: click one to rename it, resize it or delete it, then save. The form now fills in on any computer.

Fill in my details

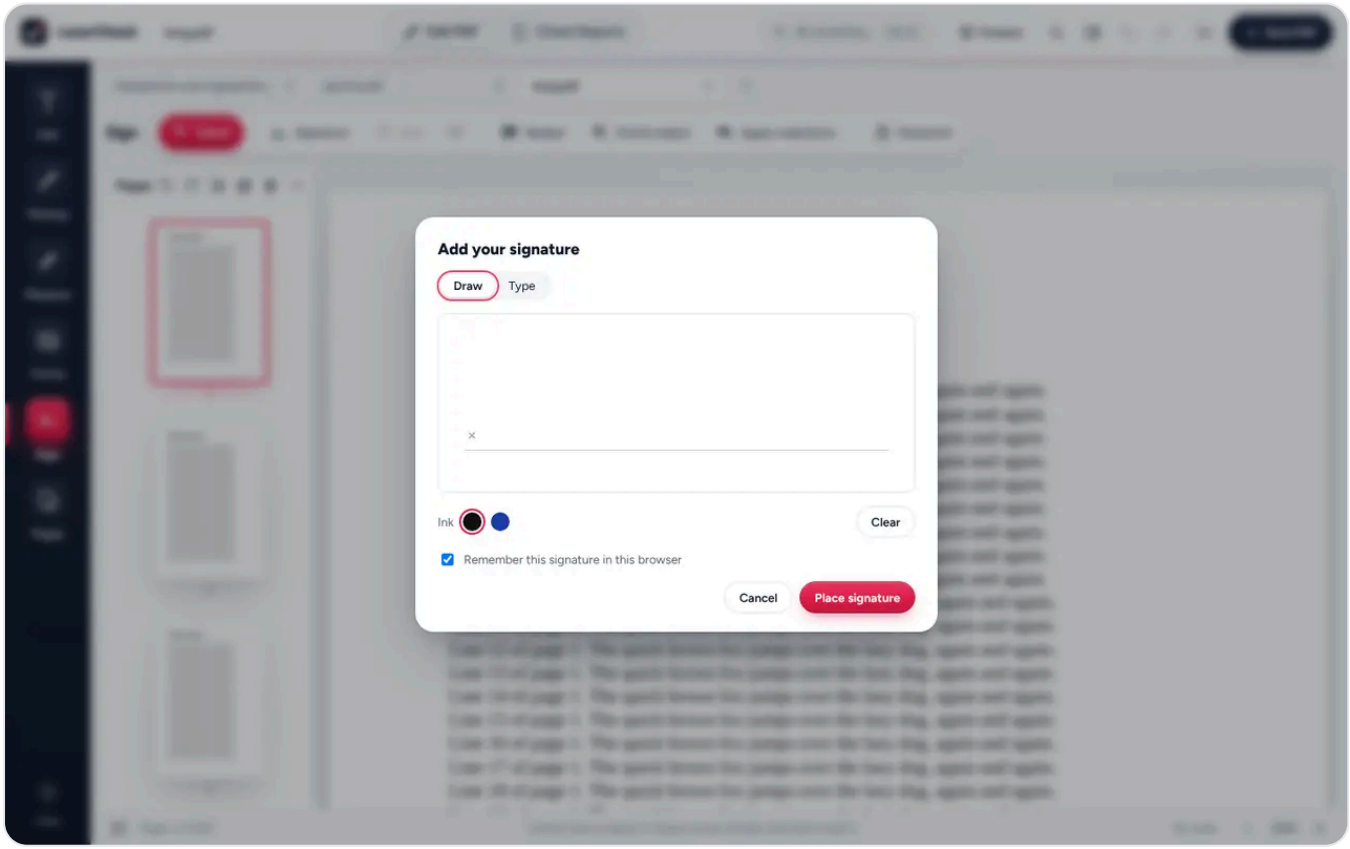
Save your name, company, address, phone, email and so on once. **Fill in my details** then recognises matching fields on any form (it understands different wordings, like “Surname” and “Last name”) and fills them, with today's date in date fields. You see what will go where before it fills anything. Your details stay on this computer.

Mail merge

1. Open the form, then click **Mail merge** and choose a CSV or Excel sheet with one row per form and column headings in the first row.
2. LaserCheck matches columns to fields by name; check and adjust the matches.
3. Set how files are named, for example `{Full name} - permit`, and whether to **Flatten** the forms (make them uneditable).
4. Choose separate files (saved into a folder you pick) or **One combined PDF**.

Rows with problems, such as a value that isn't one of a dropdown's choices, are listed at the end so you can fix them.

9 Signing and security



Draw or type your signature; it can be remembered for next time.

Signatures S

In Sign mode (⌘ 5) choose **Signature**, then **Draw** it or **Type** it. Tick **Remember this signature** to reuse it. Click to place it on the page and drag to size it. Each signature is captioned with your name and the date and time it was signed, added automatically.

Sealing a document

A seal is a digital signature for the whole file. Anyone opening it in Adobe Acrobat or Reader can see who sealed it and when, and that nothing has changed since.

1. Click **Seal** (or ⌥ ⌘ S). The first time, LaserCheck creates your signing certificate from your name, organisation and email.
2. Add a **Reason** if you like, and tick **Lock the document** if no further changes should be allowed.
3. LaserCheck gets a trusted timestamp from an independent time service (DigiCert, Sectigo or GlobalSign), so the signing time can be proven. This step needs an internet connection.

See or replace your certificate under **Certificate** on the Sign toolbar (or **LaserCheck → Signing Certificate…** on Mac).

> ABOUT CERTIFICATES
>
> Your certificate is created by LaserCheck rather than bought from a certificate authority, so Acrobat shows the signature as valid but the signer as “not trusted” until the recipient chooses to trust it. The timestamp itself comes from a trusted authority.

Redaction

Redaction removes content for good: the text, pictures and drawing underneath are deleted from the file, not just covered.

1. Mark what to remove: choose **Redact** and drag over it, or use **Find & redact** (⇧ ⌘ R) to mark every match for words or phrases, and for personal details: **email addresses**, **phone numbers**, **UK postcodes**, **National Insurance numbers**, **payment card numbers** and **dates**.
2. Or use **Redact personal information…** (Edit menu, or ⌘K “redact personal”). See below.
3. Check the marks. Nothing has been removed yet.
4. Click **Apply redactions**. The marked areas become black boxes and the content underneath is deleted when you save.

> BEFORE YOU SEND
>
> Applied redactions can't be undone once the file is saved. Keep your original. To also strip the author, hidden metadata, attachments and scripts, use **Remove hidden information** in Batch.

Redacting personal information

Redact personal information… looks through every page for:

- **Names**: in fields that hold a person (Name, Engineer, Witness, Signed by, Approved by…), after titles (Mr, Mrs, Dr…), and common first names followed by a surname. Once a name is found, every other place it appears is found too.
- **Street addresses**, **postcodes**, **email addresses**, **phone numbers** and **National Insurance numbers**.
- **Bank and card details**: sort codes, account numbers, IBANs and card numbers.
- **Vehicle number plates**.
- **Faces in photos** (desktop app), found by the detection model on this computer.

With the AI pack installed (see Ask the document), you can also have the AI model read each page for names and addresses the rules miss; it takes longer. Dates are off by default.

You then see everything found, grouped, with the page. Untick anything that should stay and click a line to see it on the page. **Mark for redaction** marks the rest; check them, then **Apply redactions**. Everything is read on this computer. Rules can miss things (a name written in an unusual way, a face turned away), so look through the document before you send it.

Password protection

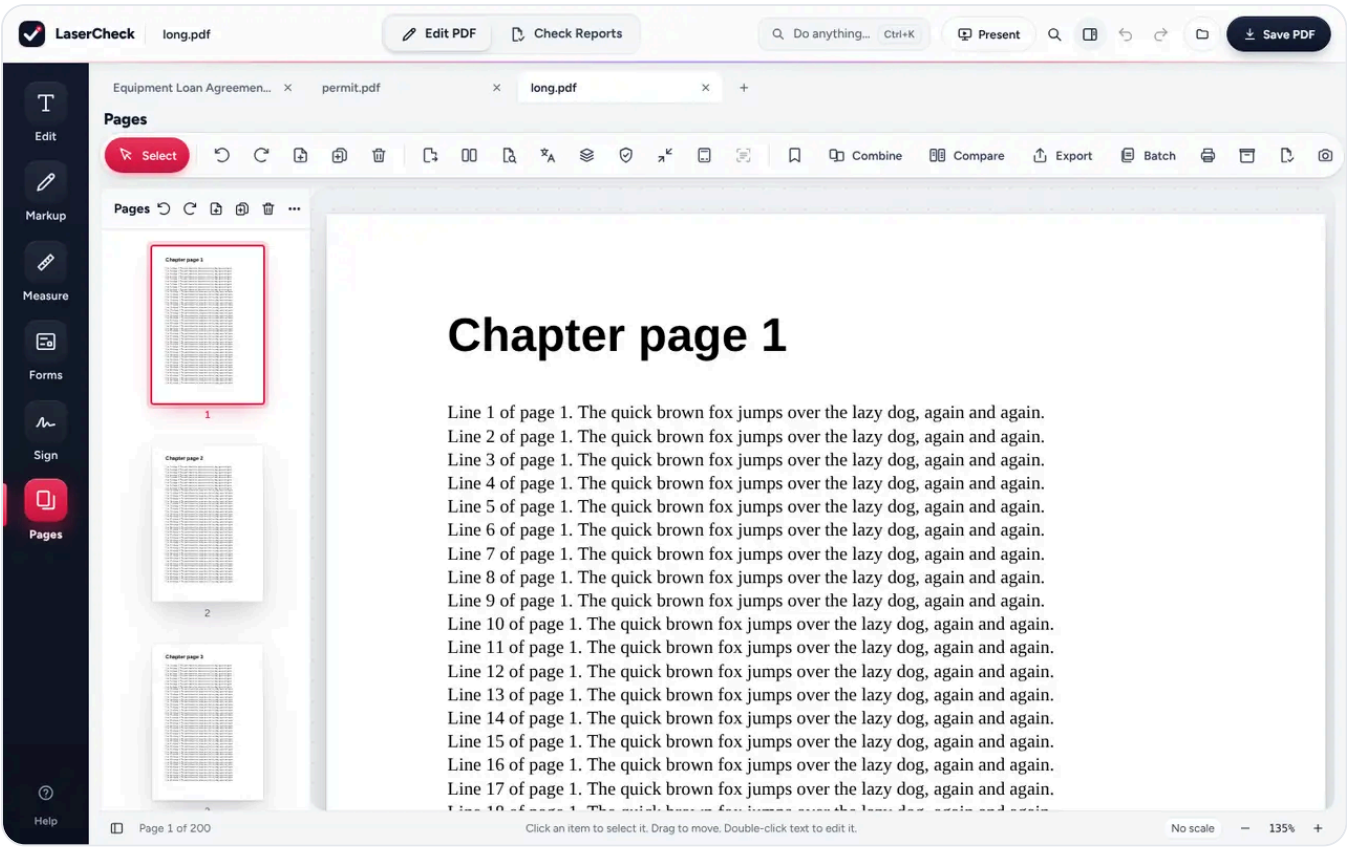
Click **Password** in Sign mode (or **File → Password Protection…**):

- **Require a password to open the document**.
- **Limit printing, copying and changes**, choosing whether to allow printing, copying, and changes (comments, forms, pages).

Files are encrypted with 256-bit AES when you save. To remove protection, open the file with its password, click **Password**, then **Remove protection**.

10 Pages and documents

Pages mode (`⌘` `6`) is for working with the document as a whole.



Pages mode: page tools, headers and numbers, bookmarks, combine, compare, export, batch, packs, checks and photo reports.

Ask the document

Click **Ask AI** on the left, under Check report, or the **Ask AI** tab in the side panel (or **View → Ask the Document**, ⌥ ⌘ A) and type a question, such as “What’s the pass criterion?” or “Who signed it?”. The buttons give a **Summary**, the **Actions and dates**, or the **Key figures**. Answers cite their pages: click a page to go to it.

It runs entirely on this computer, with a small open-source AI model (Qwen3 4B) and the llama.cpp engine. Your documents aren’t sent anywhere. The first time, LaserCheck downloads the AI pack (about 2.5 GB) and checks it before use; after that it works offline. On a Mac with Apple silicon answers take a few seconds; on other computers they can take longer.

> **CHECK WHAT MATTERS**
>
> The model answers only from the document and says when it can’t find something, but like any AI it can misread a table or miss a detail. Click the page reference and check anything important. Scanned pages need **Recognise text** first.

Feedback and feature requests

Choose **Help → Report a Problem…** or **Help → Request a Feature…**, type your message and send it. LaserCheck adds the version and your system (untick it if you prefer) and never your documents. The message opens in your mail app, so nothing goes until you send it; or click **Copy** to send it another way.

Presenting

Click **Present** at the top of the window (or **View → Present**, ⇧ ⌘ ↩) to show the document full screen, a page at a time, as it looks now with your markup. Click, → or **Space** for the next page, ← to go back, **Esc** to stop.

TOOL	KEY	WHAT IT DOES
Laser	**L**	A glowing red dot with a short trail follows the mouse.
Spotlight	**S**	Dims everything except a circle around the pointer. Scroll to make it bigger or smaller.
Pen	**D**	Draw to ring or underline something; it fades away after a few seconds. **C** clears it at once.
Zoom	**Z**	A magnifying lens under the pointer. Scroll to change how much it magnifies.
Blank screen	**B** / **W**	Black or white screen, to pause. Press again to come back.

Phone remote (desktop app): click it and scan the code with your phone's camera. Your phone becomes a remote with **Back** and **Next**, a touch pad to point with the laser, Spotlight and Blank screen. The phone and computer need to be on the same Wi-Fi; nothing from the document goes to the phone, and the remote stops when you stop presenting. The first time, your computer may ask whether LaserCheck can accept incoming connections: allow it.

Organising pages

- Select pages in the thumbnails: click, ⌘-click for several, ⇧-click for a run. Drag to reorder.
- **Rotate left / right**, **Insert blank page**, **Add pages from another PDF** and **Delete pages** are on the toolbar and the **Page** menu.
- **Extract pages** saves the selected pages as a new PDF. **Split document** saves every *N* pages as separate files.
- **Reduce file size**: **Light** (220 dpi, best quality), **Medium** (150 dpi, good for email) or **Strong** (100 dpi, smallest file).

Headers, footers and page numbers

Headers & numbers adds text to every page: a header, a footer, **Bates numbering** (prefix, start number and digits) and a **watermark** (text, strength and colour). Ready-made layouts include *Page X of Y*, *Bates number*, *DRAFT watermark* and *File name and date*. You can set text size and margin, choose pages, and skip the first page. Codes you can use: `{page}`, `{pages}`, `{date}`, `{file}`, `{title}` and `{bates}`. Changes show on the pages as you type.

Straighten pages

Scans and photos of paper often come in a degree or two crooked. **File → Straighten Pages…** (or search for it with ⌘ K) measures the tilt from the lines of text and table rules and turns the page's content back level, for the page you're on or every scanned page at once. Text, lines and pictures stay as they were, just level; Undo puts them back. Text recognised in a straightened scan should be recognised again.

Scanning pages with your phone

Choose **File → Scan with Phone…** (⌥ ⌘ N). Point your phone's camera at the QR code: it opens a page from this computer (your phone and computer need to be on the same Wi-Fi; nothing goes over the internet). Tap **Photograph a page** for each sheet, or **Choose photos instead** for ones already on the phone.

Each photo appears on the computer straightened: LaserCheck finds the edges of the sheet, squares it up (a shape near A4 becomes A4), and evens out the light so the paper is white and the writing dark. Lay the page on a darker surface for the best result. Under each page you can turn it, use the whole photo instead, or remove it. Choose **Colour**, **Black and white** or **As photographed**.

Add pages puts them after the current page (or opens them as a new document), and recognises their text so they can be searched and copied. **Add photos from this computer…** does the same with photos you already have; in the browser version, that's how scanning works. The phone's link stops working when you close the window.

Recognising text in scans

Scanned pages are pictures of text, so you can't search or copy them. **Recognise text** reads them and adds invisible text to the page, so search, copy, Edit text and Find & replace work. Use it on selected pages or all pages (**Page** menu). It runs on your computer; nothing is uploaded.

Bookmarks and links

The **Bookmarks** panel (⌥ ⌘ B) shows the document's outline. **Add bookmark** bookmarks the current page; use the arrows (or ⌥ + arrow keys) to reorder and nest them, F2 to rename, and ⌫ to delete.

Combine

Combine joins PDFs and pictures into one document in the order you choose (drag to reorder). Each file gets a bookmark, with its own bookmarks inside. For a structured package with a cover and contents, use Build a pack instead.

Compare

Compare shows two versions of a document side by side, with removed wording highlighted on the left and new wording on the right. Compare against another open document or a file; put the older version first.

Drawings: overlay

For drawings, choose **Overlay** at the top of the comparison. The new revision is lined up on the old one first (a sheet re-plotted a few millimetres over, or at a different paper size, would otherwise show as changed everywhere; the heading says how far it was moved). Then lines only in the old drawing are **red**, lines only in the new one **green**, and lines in both grey. The areas that changed are boxed and numbered, and listed on the left: click one to go to it. **Old**, **Both** and **New** flick between the two drawings and the overlay.

History and authenticity

Not sure a document is what it claims to be, such as a fit note, a certificate or a signed report? **History and authenticity** (Pages mode, or **File → History and Authenticity…**) looks for the marks editing leaves in a PDF:

- **Earlier versions.** Each time a PDF is changed and saved, many programs add the change to the end of the file, so the earlier versions are still inside. You see each save, when it was made and with what program, and which words changed. **Open** an earlier version, or **Compare with now** to see the differences side by side.
- **Signatures.** Whether anything was added after the document was digitally signed, with a button to open the version that was actually signed.
- **Details.** The program that made it and the one that last saved it (online PDF editors are named), and dates that don't add up.
- **On the pages.** Text written over other text, words in a font nothing else on the page uses (or one the file doesn't carry), text that's there but hidden, and typing, pictures or white boxes laid over a scan. Click a finding to see it on the page.

> SIGNS, NOT PROOF
>
> A genuine document can show some of these (a form typed into, a stamp added by the issuer), and a forgery that was printed and scanned again shows none of them. When it matters, check with whoever issued it.

Find blurry pictures

Choose **File → Find Blurry Pictures…** (or search for it with ⌘ K). LaserCheck checks every picture in the document and outlines the soft or very blurry ones on the page, with a list you can jump through. A small picture stretched large on the page counts as soft too. Click **Clear outlines** to remove the outlines.

In ordinary documents, **Sharpen** lays a crisper copy over a soft picture; the original stays underneath, and Undo removes the copy. Sharpening helps pictures that are slightly soft, but it can’t bring back detail lost to camera shake or focus. In **Check Reports** there’s no Sharpen: a blurry photo in a test record should be retaken, since sharpening it could look like it was altered.

Flatten

Flatten (Pages mode, or **File → Flatten…**) saves a copy where comments and markup, form fields (with their answers) and layers become part of the page: it looks the same in every viewer and can't be changed or switched off. Hidden layers are taken out completely. The document you have open isn't changed.

Remove hidden information

Before sending a file outside the company, **Remove hidden information** (Pages mode, or **File → Remove Hidden Information…**) shows what the PDF carries besides its pages, such as the author's name, the program that made it, comments, attached files, scripts, hidden layers and form fields, and saves a clean copy without the ones you tick. Saving never includes earlier versions of the file or anything deleted from it.

Convert to Word or Excel

File → Convert to Word… or **Convert to Excel…**, or **PDF to Word** and **PDF to Excel** on the start screen. Everything runs on your computer: nothing is uploaded.

- **Word, same layout**: each page comes out the same size and looks like the PDF. The text sits where it was, in the same font, size, bold, italic and colour, with the same line breaks, and you can edit it in place. Photos and logos are pictures you can move or replace. Lines, boxes, shading and drawings are kept as they look, behind the text.
- **Excel workbook**: tables come out cell for cell, with merged cells, borders, shading and bold kept. Numbers stay numbers, shown as they were written (1,240 · 4.0 · 15%), so you can add them up; dates and reference numbers stay exactly as written. Columns set out without lines are split into columns too, and “Label: value” headers become a label column and a value column. Choose **a sheet for each page**, **everything on one sheet**, or **only the tables**.

HOW CLOSE IS IT?

Word and PDF work differently, so a few things can't carry over exactly. Fonts that aren't on the computer opening the file are replaced by the nearest one, so line lengths can shift slightly. Each block of text is its own text box: edit a few words freely, but for a heavy rewrite, **Word, text only** gives flowing paragraphs instead. Lines and shading can't be edited as Word shapes. Scanned pages need **Recognise text** first to give you words; without it they come out as pictures.

Translate

Translate (Pages mode, or **File → Translate…**) makes a copy of the document in another language, laid out like the original: the same page size, tables, lines, pictures and colours, with each paragraph translated and set back in its place. The text is re-wrapped to fit and made a little smaller where the translation runs longer. Choose the pages, and what to make:

- **A translated PDF**, opened in a new tab.
- **A translated Word document**, laid out the same way as **Convert to Word**.
- **Side by side**: each original paragraph beside its translation, page by page, so someone who reads both languages can check the wording. **Copy all as text** copies them in order.

The language is detected for you; change it if it's wrong. English translates to and from more than 50 languages, including German, Dutch, Danish, Polish, French, Spanish, Italian, Portuguese, Swedish and Norwegian. For languages in other scripts (Arabic, Hebrew, Hindi, Thai, Chinese, Japanese, Korean and others) the bundled fonts can't set a PDF, so translating into them makes the Word document and the side-by-side view; translating from them works in full. Between two other languages (Danish to German, say) it goes through English. Numbers, units, dates, serial numbers, cable IDs and document codes are kept exactly as written, so 500V, ≥ 1 GΩ/km, 26/09/2025 and TP-014 come through unchanged.

YOUR DOCUMENTS STAY ON YOUR COMPUTER

Translation runs inside LaserCheck, with Mozilla's Firefox Translations engine. The only thing downloaded is a language pack (20 to 40 MB for each pair of languages, each way), once, from Mozilla; nothing from the document is sent with it. Packs are kept on your computer and work offline after that. **Language packs…** in the Translate window shows what's installed and lets you delete packs. In LaserCheck Lite, Mozilla's server won't let a web page download the pack itself, so the window gives you the files to download and add.

MACHINE TRANSLATION

It reads well for procedures, reports and certificates, but it can get a technical term or a negative wrong. Use the side-by-side view, and have wording that matters checked by someone who reads the language. Scanned pages need **Recognise text** first.

Export

Export (⌥ ⌘ E) also offers:

- **Word, text only**: words, paragraphs, headings and bold, to reflow and restyle. Pages with no text (drawings, unrecognised scans) go in as pictures.
- **Plain text**: just the words.
- **Pictures**, one per page: **PNG** (sharp, best for drawings and text) or **JPEG** (smaller, best for photos and scans), at a picture quality you choose.

For a single table, the **Table** tool on the Edit toolbar lets you drag round it and check the grid before saving it to Excel.

Name and file PDFs

File → Name and File PDFs… makes copies of PDFs, named from what's inside them. **Your files stay exactly as they are**: nothing is moved, renamed or changed.

1. Click **Choose PDFs…** or **Choose a folder…** (folders inside it are included). LaserCheck reads the first page of each.
2. Check the suggested names. A **report** (a document with a cable or circuit in its header) is named from its cable, test, end and date, such as "Z01-1 Insulation Resistance Test End A 2026-09-01"; other documents from their title and date. Change the patterns with `{cable}`, `{location}`, `{test}`, `{end}`, `{date}`, `{project}`, `{title}` and `{name}` (the current name), or type over any single name.
3. Choose **Folders**: all in one, or by project, cable or test. Documents that aren't reports go in "Other documents".
4. Read the notes: a file named for a different cable than it's about, the same report sent twice, the exact same file twice (left out), or a scan with no text (its name is kept: recognise its text first to name it).
5. Click **Copy** and choose where the copies go. A file already there is never overwritten: the copy gets a number added. A log of every copy ("LaserCheck filing log …csv") is saved in that folder.

In the browser version the copies are downloaded as a zip, in their folders.

Batch processing

Batch runs the same steps on many PDFs at once. Add the files, tick the steps, and choose the output (**PDF**, **Word**, **Excel** or **Text**) and what to add to the file names:

- Page numbers (Page 1 of 9)
- Bates numbers, continuing across the files
- Watermark

- Find & redact, including the personal-details patterns
- Recognise text in scanned pages
- Remove hidden information (author, metadata, attachments, scripts)
- Reduce file size (150 dpi pictures)
- Password to open

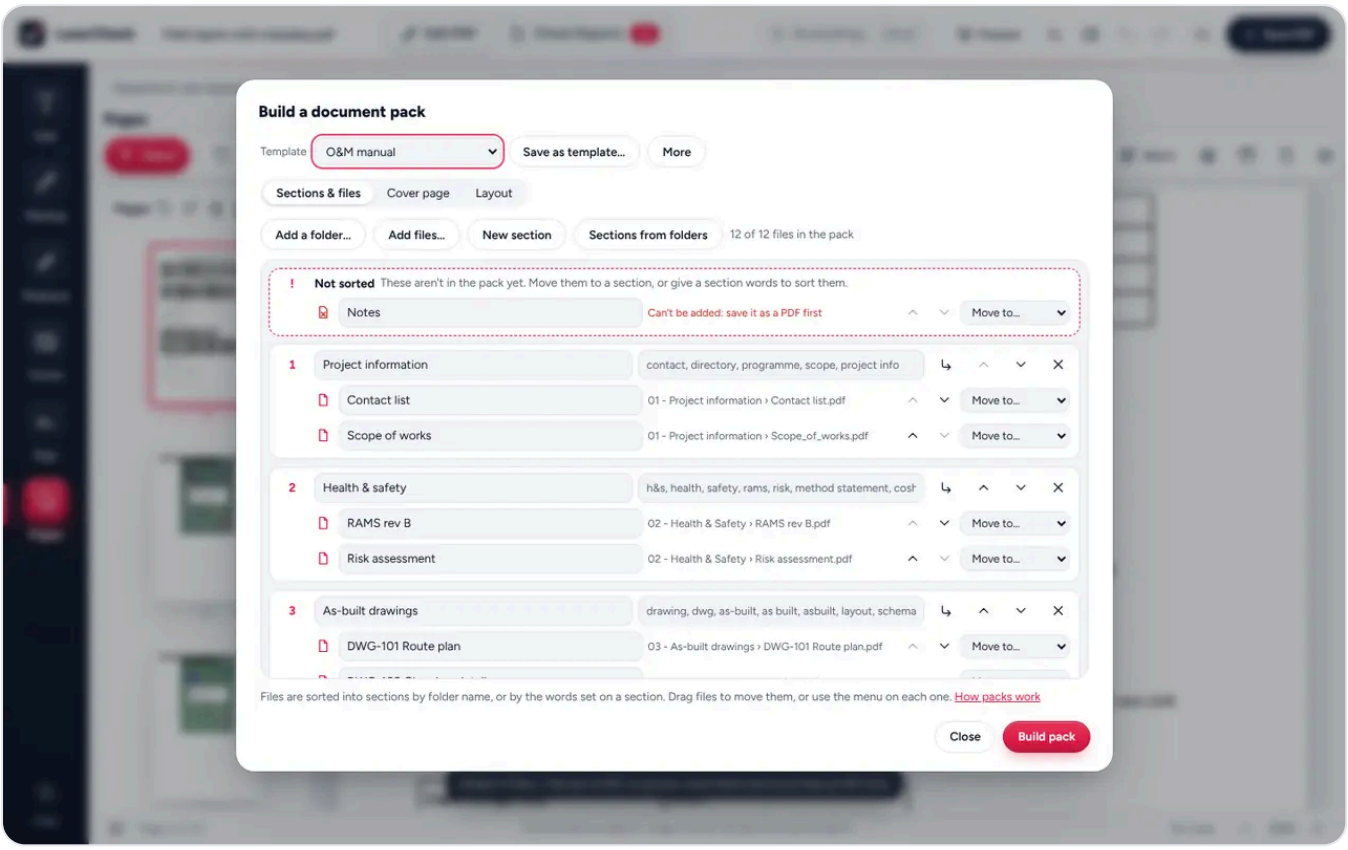
Site photo report

Photo report (start screen, Pages toolbar, or **File → New Site Photo Report…**) turns site photos into a tidy A4 report. Add a title, site or job, and who prepared it, then add photos. LaserCheck reads each photo's date, time, GPS location and compass direction. Give each photo a caption, drag to reorder or **Sort by time taken**, and choose 1, 2 or 4 photos per page. Each photo can show when and where it was taken, with a QR code linking to its location on a map.

> PHOTO FORMATS
>
> JPEG and PNG work. iPhone HEIC photos aren't supported: set the camera to **Most Compatible**, or export the photos as JPEG.

11 Document packs

Turn a job folder into a finished handover pack, O&M manual or submission: cover page, linked contents, numbered sections with divider pages, page numbers and bookmarks, in one step.



A job folder sorted into an O&M manual's sections. Anything that doesn't match waits under Not sorted.

Building a pack

1. Open **Build a pack** from the start screen, the Pages toolbar, or **File → Build a Document Pack...** (⇧ ⌘ B).
2. Pick a **Template**: *O&M manual*, *Fibre handover pack*, *Tender submission*, *Start from my folders*, or one of your own.
3. Click **Add a folder...** and choose the job folder (or drag the folder onto the window). Files are sorted into sections.
4. Check the sorting. Rename titles, drag files between sections, or use the **Move to...** menu on each file. Add sections and sub-sections as needed.
5. Fill in the **Cover page** tab and adjust the **Layout** tab, then click **Build pack**. The pack opens in a new tab, ready to check and save.

How files are sorted

- **By folder name**: a file in a folder called “03 - As-built drawings” goes to the section called “As-built drawings”. Numbers at the start are ignored.
- **By words**: each section has a box of words (for example *otdr, light meter, test*). A file whose name or folder contains any of them goes there.

- **Sections from folders** replaces the sections with your folder structure: top-level folders become sections, the next level sub-sections.

What goes in the pack

- **Cover page**: title, subtitle, your logo, and details such as project, client, site, document number, revision, date and prepared by. Use `{today}` for today's date and `{me}` for your name.
- **Contents page**, listing sections, sections and sub-sections, or every document. Every line is a clickable link.
- **Divider pages** for sections (and optionally sub-sections), each listing what's inside with page numbers.
- **Page numbers** ("Page 3 of 120" or "3") and a **header** on every page except the cover. The header can use `{title}`, `{section}` and any cover detail, like `{Project}`. Tick **White box behind them** to keep them readable on drawings. They stay upright on rotated and large drawings.
- **Bookmarks** for every section and document, keeping each document's own bookmarks and internal links.

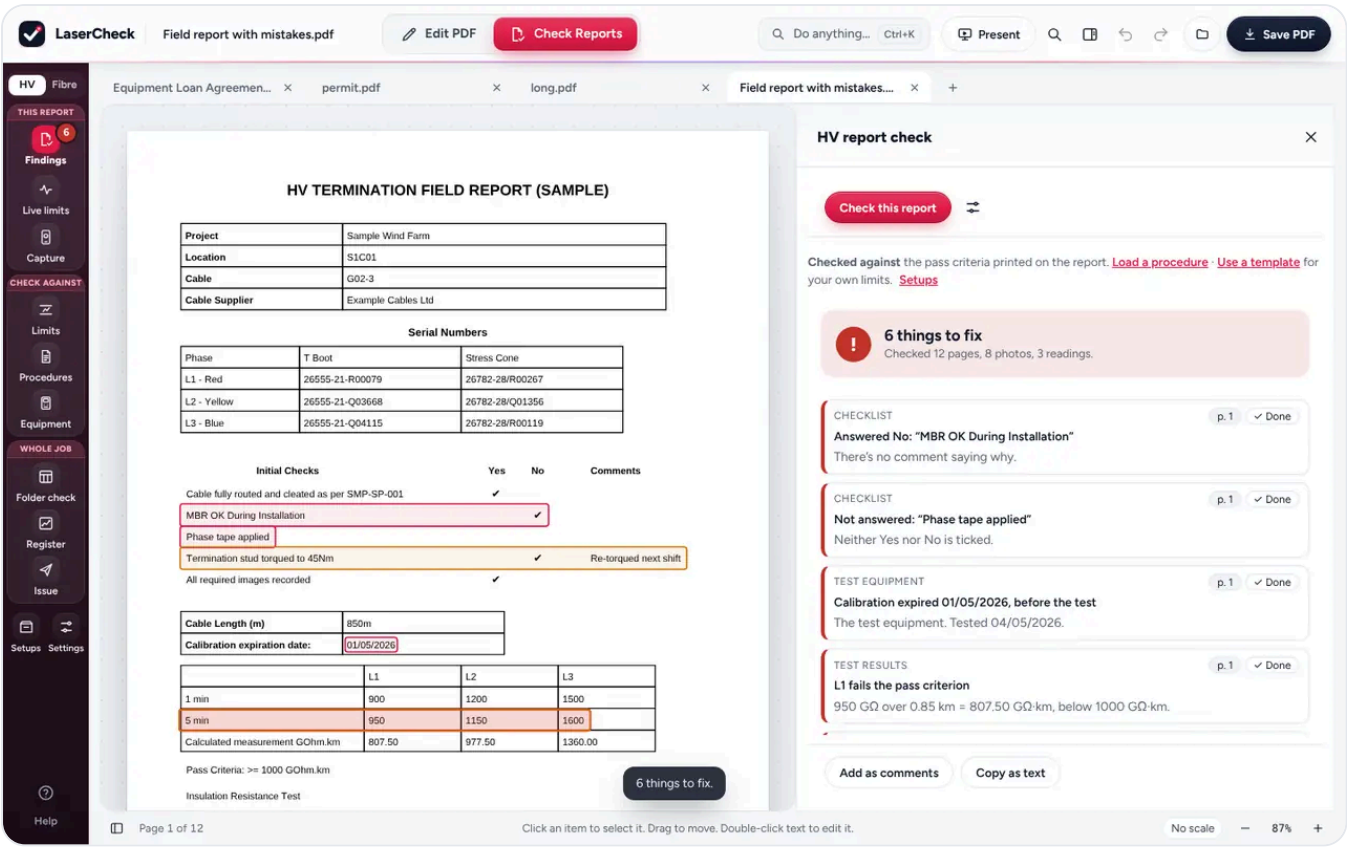
Templates

Save as template… keeps the sections, sorting words, cover details, logo and layout, so the next job takes one click: choose the template, add the folder, build. **More → Export template file…** saves it as a file to share with colleagues, who use **Import template file…**.

> GOOD TO KNOW
>
> - Word and Excel files can't go in directly: save them as PDF first. They're shown in red.
> - Password-protected PDFs ask for their password as the pack is built.
> - Fillable forms inside included PDFs keep their filled-in look but aren't fillable in the pack.
> - The dialog remembers your files until you close the app, so you can tweak and build again.

12 Report checker

Reads a field report the way a fussy client would, before it goes out, and points you to anything that needs a second look.



Findings grouped as Problems, Worth checking and Looks right. Click one to jump to the page.

Running a check

Click **Check Reports** at the top of the window, or **Choose a report…** on the start screen to open a report and check it in one go. When you open something that looks like a field report, LaserCheck offers to check it; choose **Don't ask again** to stop that. You can also use **Tools → Check Report Before Sending** (⇧ ⌘ K). Results appear in the **Checks** panel, starting with a verdict: **Ready to send**, or how many things there are **to fix**. Only clear problems are listed there (a reading outside its limit, a blank detail, the wrong cable in the file name, a photo used twice), each with a **Fix** button where the fix is clerical. Click a finding to jump to its page and highlight the spot. Under them, **What checks out** shows what passed: readings within their limits, test equipment in calibration, photos new and in order, details consistent across the pages. The Findings button (and Check Reports, while you're editing) shows the number to fix, or a green tick.

Some findings need a person's judgement rather than being wrong: two signatures at the same minute, a reading falling between 1 and 10 minutes, a stamp the app may have misread. They're kept out of the way: tick **Show all checks** to see them, with everything that passed. A reviewer may want it on; it's remembered. ✓ **Done** takes a finding off the list (it stays off when you check again), and pictures that are part of the report's template, such as a form's diagrams or its example photo, are recognised by being in the same place on every report and aren't treated as photos. **Add as comments** puts each issue on its page as a comment; **Copy as text** gives you a list for an email. A pack containing several reports is checked report by report.

Check Reports is a workspace of its own. The rail down the left holds the checker's tools instead of the editing modes, the page thumbnails and editing tools step aside, and the findings get a wide panel beside the page.

At the top of the rail, **HV** or **Fibre** says which kind of report you're checking. It follows the report you checked last (a fibre report switches it to Fibre by itself), and you can switch it by hand, for example to set up a fibre spec before opening a fibre report. The rail then shows only the tools for that kind, in groups:

	HV REPORTS	FIBRE REPORTS
This report	**Findings**; **Live limits**: every reading coloured on the page (see Live limits); **Capture**: a shot list on your phone (see Site capture)	**Findings**, with the fibre-by-wavelength grid
Check against	**Limits**, **Procedures**, **Equipment**	**Fibre spec** (see Fibre reports), **Equipment**
Whole job	**Folder check**: which reports a job is missing (see below); **Register**: every reading with charts and Excel (see Results register); **Issue** (see Issue to client)	**Issue**

Setups, at the bottom beside Settings, keeps what reports are checked against (the equipment register, limits and limit sets, procedures, fibre specs and which checks run) as named setups, one per client or job. **Save setup** keeps what's set up now under a name; **Load** switches to a saved one; **Save to a file…** writes a `.lpsetup` file and **Open a setup file…** reads one, so a setup can be backed up or shared with the team through a shared folder (OneDrive, SharePoint or Teams); **Clear the setup** starts afresh. Loading or clearing asks first if what's set up now isn't saved. The setup in use is named on the **Checked against** line. Give a setup the words its reports mention (a client name, a job number) in **Offer it for reports that mention…**, and when a report mentioning them is checked while another setup is in use, a bar above the findings offers to switch: **Switch and check again**, or **Not now**. It never switches by itself. Setups don't include the history of reports checked before, and nothing is sent online.

Settings, at the bottom, chooses which checks run and holds the **History** of photos and equipment remembered from reports checked before. At the top of the findings, **Checked against** says what the report was measured against (the criteria printed on it, your procedures and limits, or your fibre spec), with a link to change it.

Click **Edit PDF** at the top to go back to editing; Check Reports brings the findings back without checking again. The checker reads the page as it will be saved, so corrections you've made (and form fields you've filled in) are what get checked.

Seeing findings on the page

While you're in Check Reports, everything the check flags that has a place on the page is outlined there: **red** for things to fix, **amber** for things worth a look, for HV and fibre reports alike. Click a finding in the list and the page moves to bring that spot to the middle of the screen, picked out with a stronger outline (and its

card marked in the list) until you click another. Findings marked Done lose their outline; back in Edit PDF, the outlines aren't shown.

Fixing what it finds

Some findings have a **Fix** button: the clerical ones, where the right answer is clear.

Review fixes: when any can be fixed for you, a bar above the findings says how many. **Review fixes** lists them together, each as before → after: the file name, a blank detail (with the value used elsewhere in the report filled in for you), a detail spelt two ways (pick the right one), a serial number or calibration date from your equipment register. Ones with a clear answer start ticked; a recalculated result such as a per-km figure is offered unticked, for you to decide. **Apply** makes them all in the report's own font, and one Undo takes them all back. Readings, Pass/Fail, ticks and signatures are never in the list: they're what the engineer recorded.

- **A blank detail** ("Cable Supplier is blank"): type it, and it's added in the same column and font as the other details.
- **A detail written two ways** (the project on page 1 and page 2): choose which is right, and the other is changed to match.
- **The file name names the wrong cable**: the document is renamed.
- **Your equipment register**: a serial number one or two characters off a registered one, or a model or calibration due date that disagrees with the register, is changed to the register's (a date is written the same way as the report's).
- **Per-km figures that divide by the length**: recalculated as reading × length.

You see the change before it's made. It replaces the text on the page, in the same font, the way **Edit text** does, and ⌘ Z undoes it. The finding is marked done; **Check this report** again to confirm. Test readings, ticks, Pass/Fail and signatures are never changed for you: they are what the engineer recorded, so those findings are for a person to sort out.

Live limits

Click **Live limits** on the left. Every reading the checker can test against a limit is outlined on the page: **green** within it, **red** outside it (with a ✕), **amber** when it can't be told (">5" against a minimum of 10). Hover over one to see the limit. The limits are the form's own printed pass criteria, insulation resistance against its criterion (per km where the form asks), your procedure limits and the limits kept from your procedures.

It updates as the page changes: correct a reading with Edit text, type into a form field, or add text, and its colour follows. A bar at the bottom counts the readings within and outside; click its ✕, or Live limits again, to turn it off. The outlines are on screen only: they are never saved in the PDF.

What it checks

AREA	CHECKS
Details	The cable and location in the header against the file name and the stamp on every photo; blank fields; a field filled in differently on different pages.
Photos	The same picture used twice (even re-saved or re-stamped); captions with no photo.
Photos used before	The same photo in two reports of one document (a pack), or in a report you checked before. See *History* below.
Photo times	Photos taken after sign-off, in the future, or on another day from the report's date; "as left" before "as found"; photos spread over several days; timed readings out of step (an "@ 5 min" photo before its "@ 1 min" photo, or far later than the test takes); many photos with the same stamp; photos outside your working hours, if you set them.
People in photos	Faces and bodies in photos. Hands, gloves and boots are fine. Desktop app.
Checklists	Items not answered, both Yes and No ticked, or answered No without a comment.
Serial numbers	A value written differently from the rest of its column; the same number on two rows; the table against the label in the photo.
Test results	Insulation resistance: pass/fail against the criterion printed on the form, per-km arithmetic, units, readings that fall between 1 and 5 minutes, readings written as text.
Procedure limits	Every reading against the limits you load from your procedures. See below.
Test equipment	Calibration expired at the test date or close to it, for every instrument listed; a missing serial or calibration date; the same serial with two models or two calibration dates across reports; a serial one or two characters off one used before; your equipment register.
Signatures	Signature blocks without a name; signatures with identical timestamps.

Choosing which checks run

Click **Settings** on the left in Check Reports (or the settings button next to **Check this report**). The **Checks** tab switches groups of checks on or off; **Photo times** sets how many days apart photos can be, your working hours (leave blank for shift work) and how much leeway timed readings get. Your choices apply to every report.

Is the value right?

Three checks look at the numbers themselves:

- **Pass criteria on the form.** When a form prints its own criteria, such as “Gland Plate to external main earth ≤ 2.0 mΩ”, every value in the column with that heading is checked against it. A value that fails is a *Problem*, even if the report says Pass.
- **Phase balance.** On one cable the three phases are usually similar. A phase ten or more times lower than the other two is flagged, even when it passes, because it can point to moisture, contamination or a test fault.
- **Wrong unit.** A reading about a thousand times off the other phases, such as 1.25 among readings of 950 and 1100 GΩ, is flagged as probably read in another unit (here TΩ), with what it would be in the table's unit. Values written with their own unit are left alone.
- **Pass / Fail.** Where a report writes Pass or Fail, a written “Pass” alongside a value that fails is a *Problem*, and so is any “Fail”, so a failed test can’t slip out unnoticed.
- **Leakage against resistance.** On insulation resistance tests, the test voltage divided by the leakage current should give the resistance reading. When the two don’t agree (allowing for how precisely each is written, decimal commas included), one of them was probably typed wrongly.
- **Unusual values.** Once the checker has seen the same test in at least four earlier reports (see *History*), a reading far outside everything it has seen is flagged as worth checking. It stays quiet until it has enough history, and it only flags values well outside the usual spread.

Checking every report in a folder

Click **Folder check** on the left in Check Reports (or **Tools → Check Every Report in a Folder…**) and choose a folder. Every PDF in it and in its subfolders is checked, HV and fibre reports alike, against the setup in use; drawings, letters and other PDFs that aren’t reports are skipped. Tick **Read the date stamps on photos** to include the photo checks (slower). Afterwards the folder’s reports stay listed down the left of Check Reports, by subfolder, each marked Ready or with how many things it has to fix, and the first report to fix opens. Click any report to open it with its findings (straight away: it isn’t checked again), or use **‹ ›** and **Next to fix** to step through; they all open in the same tab. **Summary** shows how many were found, checked and skipped (and why), the fibre cable grids, and **Save as CSV…**. Close the list with ✕; the Folder check window can bring the last folder back.

Fibre reports are also gathered by cable into one fibre-by-wavelength grid, so a job with one iOLM report per fibre shows the whole cable: fibres **missing** from the fibre spec’s list, fibres **tested twice** (with the files), and each fibre’s loss and anything else that failed.

Checking a whole job for missing reports

In the Folder check window, click **Missing and doubled tests (HV)…** (or **Tools → Check a Whole Job for Missing Reports**) and choose the job’s folder, or a set of reports. LaserCheck reads the first page of each (cable number and test type) and shows a grid of cables against test types. A test is expected for a cable when most cables in the same group (B02-…, H02-…) have it, as many times as they usually do, so an earth verification at both ends counts as two. Missing reports are shown in red, and reports sent more than once in amber. **Copy the list** gives you the gaps as text for an email. Nothing is changed and nothing leaves your computer.

Meter readings

For each test photo that matches a reading in the report's table (for example "L1 @ 1 min", or a photo captioned with a column heading like "Steel Tube to internal main earth"), the checker crops the instrument's display and shows it beside the value the table records. Look down the list: a display that shows a different number, or something other than the reading, is easy to spot. The crops appear only in the Checks panel; they are never added to the PDF, and **Add as comments** leaves them out.

Checks from your procedures

LaserCheck can read a procedure, method statement or RAMS and draft checks from it. Click **Procedures** on the left in Check Reports and choose **Add a procedure…**. The document is read on this computer; nothing is sent anywhere, so restricted and confidential procedures are fine.

It looks for:

- **Steps that need a photo or data**: task-plan steps marked with the camera symbol (it learns the symbols from the document's own symbol key), steps that say to photograph something, and "picture points … as a minimum" lists.
- **Limits written in the text**, such as "a minimum of 12m required for termination". These are off until you tick them and say which reading in your reports they apply to.
- **Hold and witness points** (in task plans, and lines starting "Hold point" or "Witness point"), listed for information.
- The procedure's **numbers and revision**, its **project**, and the **reports** its steps are recorded on (from the "documents to be completed" column).

Review the drafts: untick anything you don't want, and change **Look for** to the words your reports use for that photo or record. Then **Keep these checks**.

From then on, every report of one of those types that mentions the project (or quotes the procedure number) is checked against it. You'll see when something the procedure asks for can't be found on any page of the report, readings outside the limits you kept, and a report that quotes an older revision of the procedure. Matching is by words, so treat a "nothing found" as a prompt to look, and mark it done if it's there under different wording.

A new revision of a procedure

When a procedure is revised, the question is what changed. Under **Procedures**:

- **Compare two revisions…**: choose the old and the new PDF (in either order; they're put in revision order).
- **New revision…** on a procedure you already check against: choose the new PDF. Adding a procedure with the same document number as one you have, but a different revision, does the same.

The comparison lists the **limits and test values** that changed (old value struck through, new in green), limits added or dropped, **hold and witness points** added, removed or changed, the **photos and records** it asks for, and every sentence whose **wording** changed, with the words marked. Page headers and footers are left out. **Copy as text** gives you the list for a briefing or an email.

For a procedure you check against, **Move my checks to revision …** moves your check set to the new revision: what you'd switched on or off and the reading labels you set are kept, and the new revision's values are taken. Review the list and save. **Keep both separately** keeps the old set and adds the new one.

Both PDFs are read on this computer; nothing is sent anywhere.

Results register

Click **Register** on the left in Check Reports (or **Tools → Results Register…**) and choose a job's reports, a folder of them, or this document. Every reading in them is listed: the cable, the test, the reading (such as "5 min L1"), the value and its unit, and whether it's within its limit, using the same limits as the checks (the form's own criteria, your limit sets and procedures).

You see the totals, the readings **outside their limits** (with the limit and the page), and a chart for each kind of reading across the cables: a dot per cable, red outside its limit, green within, grey where no limit applies. Hover over a dot for its value. Reports where no readings were found are named, so nothing is quietly missed.

Save as Excel… gives three sheets: every reading, a summary per reading (lowest, median, highest, how many outside), and the readings outside their limits. The reports themselves aren't changed.

Site capture

Get the photos and readings right while you're still on site. Click **Capture** on the left in Check Reports, or **File → Site Capture…**, with the report open (a blank one is fine).

1. Choose what goes on the list: the **readings left blank** in the report's tables (each blank under L1, L2, L3 or Red, Yellow, Blue becomes one reading, with its unit from the table), the **photos your procedures ask for**, and any more photos or readings you type. Check the text stamped on each photo (the cable and location are filled in from the report).
2. In the desktop app, point your phone's camera at the QR code. The list opens on the phone (same Wi-Fi; nothing goes over the internet). Tap the camera beside each item. In the browser, use **Add photo…** beside each item instead.
3. Each photo is checked as it arrives. It's sent back, with the reason and a **Retake** button, if it's **blurred**, **too dark** or washed out by glare, **too small**, the **same photo** as another on the list or one in a report checked before, **taken before the capture started** (an old one from the camera roll) or, in the desktop app, has **someone in it**. **Keep it anyway** accepts it, and the photo record says what it was kept despite.

4. For a reading, photograph the meter's display. LaserCheck tries to read the big seven-segment digits and puts what it read in a box on the phone. **Check it against the display**, correct it if needed and tap **Confirm**; if it couldn't read the display, type the reading. Nothing is filled in until you've confirmed it.
5. Click **Add to the report**. Confirmed readings are typed into their blank cells, in the report's own font, and the photos are added at the end as a **photo record**: two to a page, each stamped with the date and time it was taken and your text, captioned with the item, the reading and whether it was read from the display, corrected or typed.

The display reader is made for the large digits of an insulation tester or ohmmeter, photographed square-on and filling most of the frame. Glare, an angle, or a display that's mostly symbols makes it give up, and you type the reading instead. It never guesses quietly: whatever it reads is shown to you to confirm.

Issue to client

The last step before a document goes out. Click **Issue** on the left in Check Reports, or **File → Issue to Client...**.

1. The document is checked (or the last check is used). If problems are still open, they're listed: **Review them first**, or, to issue anyway, type why. The reason is printed on the transmittal, so nothing goes out unexplained.
2. Fill in the **transmittal**: who it's to and from, the project, the transmittal number (numbered on from the last one each day), the purpose (for information, review, approval, construction, as built or record), the revision and any notes. Your details are remembered.
3. Choose what the issued copy gets: the transmittal sheet on the front (with the reports inside and their pages, and what the check found); markup and form answers made part of the page; hidden information removed; and, in the desktop app with a signing certificate, a digital seal with an independent timestamp.
4. Click **Issue...** and save. The file is named with the transmittal number and revision. The document you have open isn't changed.

What reports are checked against

Nothing is loaded when LaserCheck is installed: no limits, procedures or register. Until you add them, readings are only checked against criteria printed on the report itself (such as "Pass Criteria: ≥ 1000 GΩ·km"). The top of the findings says what's in use: when nothing is, it offers **Load a procedure...**, **Use a template...** and **Equipment register...**; afterwards it lists what reports are being checked against. Live limits offers the same when it has nothing to check.

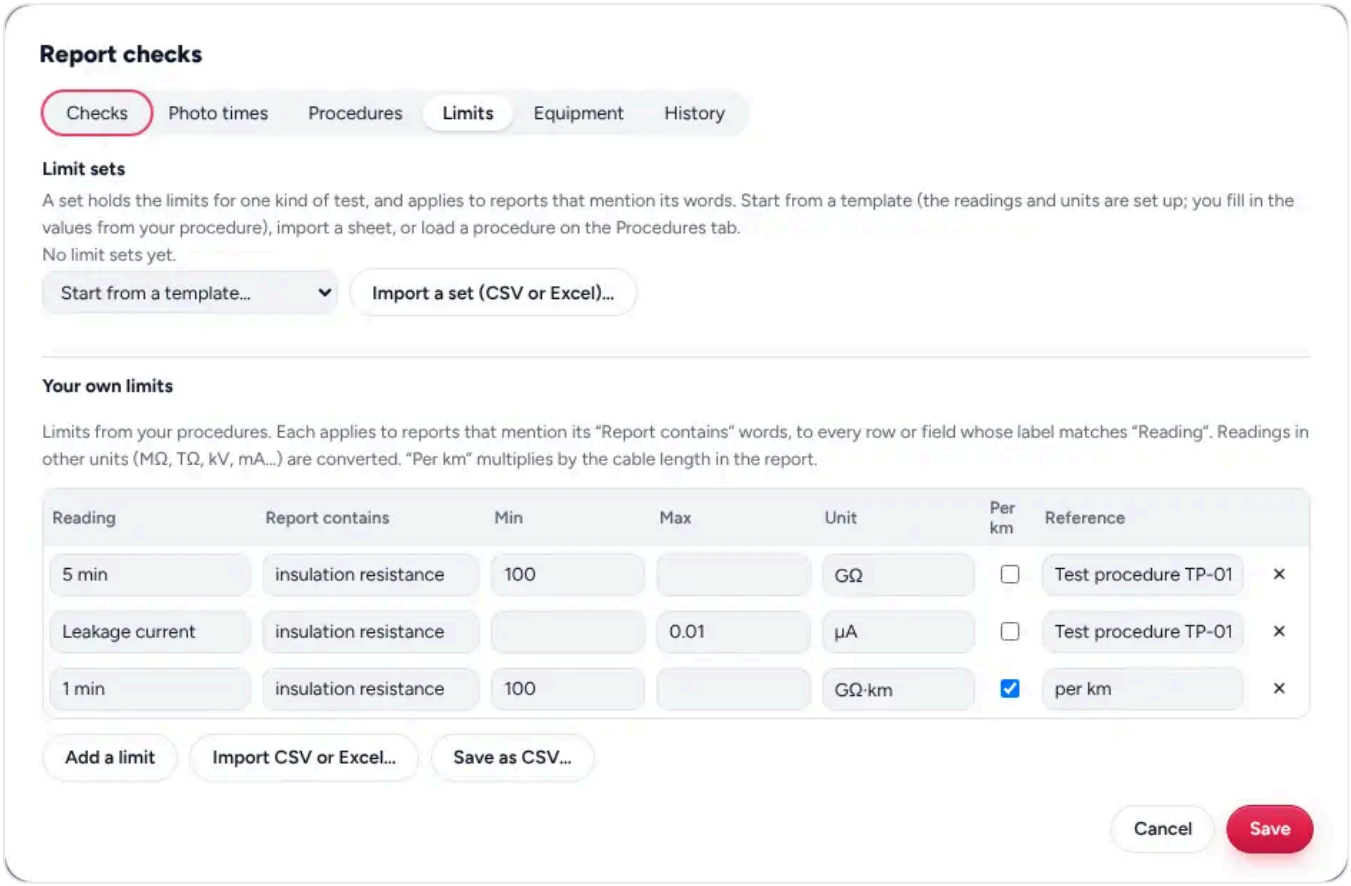
Limit sets and templates

A limit set holds the limits for one kind of test, and applies to reports that mention its words. Click **Limits** on the left in Check Reports, then:

- **Start from a template…**: insulation resistance, sheath test, HV/VLF withstand, contact resistance (DLRO), earth bonding and continuity, torque, or fibre splice and OTDR. The template sets up the readings and units; **the values are blank**, for you to fill in from your procedure. A set with no values checks nothing, and says so.
- **Import a set (CSV or Excel)…**: a sheet in the same columns as below. A colleague's set, saved with **CSV**, comes in this way.

When you fill in **Reading**, the box suggests the labels of rows and fields with numbers in the report you have open, so you can pick the report's own wording. Untick a set to stop using it without deleting it. Limits from procedures you load work the same way and are kept with the procedure.

Procedure limits



Procedure limits: one row per limit, imported from a spreadsheet or typed in.

Under **Your own limits** (below the sets), add one row per limit from your procedures, or import them from a CSV or Excel sheet with these columns:

COLUMN	WHAT TO PUT	EXAMPLE
Reading	The label of the row or field the reading is on. Matched as whole words, so "5 min" doesn't match "15 min".	5 min

COLUMN	WHAT TO PUT	EXAMPLE
Report contains	Words the report must contain for the limit to apply. Blank applies it to every report.	insulation resistance
Min, Max	Either or both.	100
Unit	The limit's unit. Readings in other units are converted: MΩ, GΩ and TΩ (and "1T"), mA and µA, kV, km.	GΩ
Per km	Yes to multiply each reading by the cable length in the report before comparing.	yes
Reference	Shown with the finding, so the reviewer knows where the limit comes from.	TP-014 6.2

A reading outside a limit is a *Problem*. ">1000" counts as at least 1000. If a limit applies to a report but no matching row is found, you get a note, so you can adjust the label. **Save as CSV** exports the table to share with colleagues.

Test equipment and your register

The checker reads each instrument from the report: "Test Equipment", "Serial number" and "Calibration expiration date" fields, or a table with serial and calibration columns. The test date is when the form was signed, or the date in its header. If a report gives only the calibration date, the due date is worked out from **Calibration lasts** (12 months unless you change it).

Under **Equipment** is your **equipment register**, kept in LaserCheck: one row per instrument with its model, serial number, calibration date and calibration due, and how it stands today (in date, due within a month, or expired). **Add an instrument** types one in; **Add from this report** adds the instruments the last check found, so you only fill in their dates; **Read calibration certificates…** fills rows from the certificates; **Import CSV or Excel…** brings in an existing sheet (with a serial number column and, ideally, model, calibration date and calibration due), adding to what's there; **Save as CSV…** keeps a copy. Leave Calibration due blank to use the calibration date plus the months set above it. Click ✕ to remove an instrument, then Save. Once a day, when LaserCheck starts, a note names any instruments out of calibration or due within a month, and the **Equipment** button shows how many. Each instrument is then checked against it: not on the register (and the closest serial, if it looks like a typo); out of calibration on the test date according to the register; a different due date or model. If the register shows the instrument was recalibrated after the test, the date isn't compared.

Reading calibration certificates

Read calibration certificates… builds the register from the certificates themselves. Choose one or more certificate PDFs (in the desktop app, scans and photos of certificates work too: they're read with text recognition). For each one LaserCheck finds the instrument (make and model), serial number, certificate

number, calibration date, due date and laboratory, and shows them in a table to check and correct. If a certificate doesn't give a due date, it's worked out from **Calibration lasts** and marked *estimated*.

Add to the register adds the ticked rows. An instrument already on the register is updated when the certificate is newer, and left alone when it's older. Click **Save** to keep it, and **Save register as CSV…** to share it or edit it in Excel.

History: photos and equipment from earlier reports

Each report you check is remembered on this computer: a fingerprint and a small thumbnail of each photo, and the test equipment details. Later reports are compared with them, so a photo reused from another job, or a serial number that doesn't match what was used last week, is caught. Checking a report again replaces what was kept for it; a pack, or a new version of a report, gets a single note rather than a finding per photo.

In **Settings**, under **History**, **Add past reports…** reads old reports in without checking them, so new ones can be compared from day one. **Forget this report** and **Forget everything** clear it, and you can turn remembering off.

People in photos

The desktop app looks for faces and bodies in each photo, using detection that runs on this computer; photos are never sent anywhere. Hands, gloves, boots and knees in shot are fine. A clear person is a *Problem*, a likely one is *Worth checking*. It can miss someone far away or mostly hidden, and it isn't a guarantee, so treat it as a second pair of eyes. To hide a face, use **Redact**.

Marking findings done

Click ✓ **Done** on a finding once you've dealt with it, or decided it isn't a problem. It leaves the list and stays off it for that report, even when you run the check again. Done findings are listed at the bottom, where you can **Restore** one or **Put them all back**. They're also left out of **Copy as text** and **Add as comments**.

Drawings, diagrams and logos aren't checked as photos. If one is ever treated as a photo (flagged as used twice, say), click **Not a photo** on the finding. That picture is skipped from then on, in every report.

Cut-off photos

A photo that didn't finish uploading or saving shows a flat grey block at the bottom. The checker spots these and lists them as problems, so the original can be put back before the report goes out.

What counts as a photo

Only real photos are checked for dates and repeats. Drawings, diagrams, logos, full-page scans and graphics repeated across many pages are recognised and left out; the checker lists them under *For information* so you can see what it skipped.

Reading the results

- **Problems** are almost certainly wrong and worth fixing before sending.
- **Worth checking** might be fine, but a reviewer may ask.
- **Looks right** confirms what was checked and passed.

> PHOTO STAMPS AND LABELS
>
> Tick **Read photo stamps and labels** to read the date, location and cable stamped on photos, and serial labels in photos. Text read from photos can be wrong, so a serial is only flagged when its label clearly reads a different number; labels that can't be read are listed in a single note. With photo reading on, a 50-page report takes about half a minute, and looking for people adds a few seconds.

Spelling

Every word of an HV report is looked up in a British English dictionary (with the trade's own words, such as glands, cleats and megger, added), on this computer. A word that isn't there is listed once, with the pages it's on and the likely word: *“Termnation” may be misspelt. Did you mean “Termination”?* Click **Fix**, or use **Review fixes** to put all the typos right at once in the report's own font. Where two words are close (“completd”: completed or complete?), you pick.

Names and places mid-sentence, capitals, codes and anything with a number in it are left alone. If a flagged word is right, click **Add to dictionary** and it isn't flagged again. **File → Spelling Dictionary…** lists the words you've added; they're saved with check setups, so the team can share them. Spelling findings are under **Show all checks** and in the **Review fixes** bar; they don't stop a report being ready to send.

Fibre reports (EXFO iOLM and OTDR)

Open an EXFO iOLM or bidirectional OTDR report and click **Check report**: LaserCheck recognises it and checks it as a fibre report. Under the verdict, a grid shows each fibre at each wavelength: ✓ **Pass**, or ✗ and what failed. Click a cell to go to its page.

Each event or loss budget. Above the results, **Judge losses by** switches how losses are judged, and the check runs again:

- **Each event** (the usual): every connector, splice and fibre section against its own limit. The grid shows the worst connector loss and reflectance.
- **Loss budget**: each fibre's whole loss against a budget worked out from its length × the attenuation, plus its connectors and splices (capped at the max link loss, if your spec sets one). Single events aren't judged on loss. The grid shows the loss and the budget, such as *0.42 of 0.85 dB*.

Reflectance and ORL are checked either way. The choice is remembered and saved with check setups.

Event losses are judged on the **bidirectional** (A→B and B→A averaged) figures and the instrument's bidirectional thresholds; one-way losses don't decide a pass. Reflectance and ORL are checked in each direction, since they're measured one way.

- **Every event**: connector and splice losses, reflectance and fibre attenuation against their limits, and ORL if you set a minimum.
- **The report's own verdict**: a report that says Pass with results outside the limits is flagged.
- **The instrument's settings**: the pass/fail thresholds printed on the report are compared with your spec. Settings looser than the spec let results pass that should fail.
- **Details**: a blank cable ID, a file name for a different fibre or cable, directions tested days apart, a report signed before the last test, serials against your equipment register.
- **Trace sense checks**: results that pass their limits but don't look like a healthy fibre. An event that loses 0.1 dB or more at 1550 nm than at 1310 nm looks like a bend or kink, not a poor splice; so does fibre attenuation that isn't lower at the longer wavelength. On OTDR reports, an event seen in one direction only (away from the ends) is often a ghost or a misread trace. And a fibre with more or fewer events than most of its cable is worth a look; Folder check compares this across the files of a cable too.
- **The fibres**: on OTDR reports, the link summary against each fibre's own page; fibres missing from the spec's list, repeated, or much longer or shorter than the rest.

Combined reports. When one PDF holds an HV report and fibre results, both parts are checked in one go: the HV pages against your HV limits and procedures, the fibre pages against the fibre spec. The findings come in one list, with the fibre grid and its **Judge losses by** switch above them.

The fibre spec. Choose **File → Fibre Spec…** (or **Change** above the grid) to set the limits you're held to: attenuation per wavelength, maximum splice and connector loss, maximum reflectance (a negative number, such as −50), and optionally a minimum ORL, a maximum link loss (for judging by loss budget) and the fibres the cable should have (such as 1-24). **Fill in from this report's printed thresholds** copies the limits from the open report as a starting point. Save several specs and pick the one for the job; without one, each report is checked against the thresholds printed on it.

13 Keyboard shortcuts

On Windows, use Ctrl where this guide shows ⌘, and Alt for ⌥.

Press `?` (or **Help → Keyboard Shortcuts**) to see them all in the app, with a box to find one.

Tools (single keys)

KEY	TOOL	KEY	TOOL
`V`	Select	`C`	Revision cloud

KEY	TOOL	KEY	TOOL
E	Edit text	N	Comment
T	Add text	B	Stamp
I	Image	S	Signature
W	Whiteout	M	Distance
H	Highlight	P	Path
D	Draw	G	Area
R O	Rectangle, ellipse	K	Count
L A	Line, arrow	U X	Underline, strikethrough
Q	Callout	Y J	Polygon, polyline
F	Copy formatting	?	All keyboard shortcuts
Esc	Back to Select		

Commands

SHORTCUT	DOES
⌘ O · ⌘ S · ⇧ ⌘ S	Open · Save · Save As
⌘ P	Print
⌘ T · ⌘ W · Ctrl Tab	New tab · Close tab · Next tab
⌘ 1 – 6	Edit, Markup, Measure, Forms, Sign, Pages
⌘ Z · ⇧ ⌘ Z	Undo · Redo
⌘ D · ⌫ · arrow keys	Duplicate · Delete · Nudge (⇧ for 10×)
⌘ F · ⌘ G · ⌥ ⌘ F	Find · Find next · Find & replace
⇧ ⌘ R	Find & redact
⌘ K	Do anything (command palette)
⌥ ⌘ N	Scan with phone
⌘ = · ⌘ - · ⌘ 0	Zoom in · Zoom out · Fit width

SHORTCUT	DOES
⌥ ⌘ P / C / B / Q	Show or hide Pages, Comments, Bookmarks, Quantities
⌥ ⌘ E	Export
⌥ ⌘ S	Seal document
⇧ ⌘ B	Build a document pack
⇧ ⌘ K	Check report before sending
F1	This guide

14 Licence and trial

For now, LaserCheck is free: there's no trial, nothing is locked, and nothing needs buying or activating. This section describes how licences will work when they're introduced.

Try everything free for 14 days. After that, LaserCheck still opens, shows and prints your documents, and a licence switches editing and saving back on.

The free trial

The trial starts the first time you open LaserCheck, with every feature switched on. The badge at the top right shows the days left; click it for details. When the trial ends, LaserCheck becomes **read-only**: you can open, view, search and print documents and read this guide, but not edit, mark up, sign, export or save. Nothing you set up is lost.

Licences

LICENCE	WHO IT'S FOR	COMPUTERS
Lifetime	One person. Pay once, with all future updates.	2
Yearly	One person, paid each year. If it isn't renewed, LaserCheck becomes read-only.	2
Team of 5	Five people in a company, under one key.	10
Team of 10	Ten people in a company, under one key.	20

Click the badge (or **LaserCheck → Licence…** on a Mac, **Help → Licence…** on Windows) to see the prices and buy. Payment is handled by our reseller, Lemon Squeezy, which emails you a receipt and your licence key straight away.

Activating

1. Open **Licence…** and, under **Already bought?**, enter the email address you bought with and your licence key.
2. Click **Activate**. LaserCheck checks the key online and unlocks straight away.

The licence screen then shows who it's licensed to, your licence type, and how many of your computers are using it.

Moving to another computer

Each licence works on its number of computers at once. To use it on a new one, open **Licence…** on the old computer and click **Move licence off this computer**, then activate on the new one. If the old computer has died, get in touch and we'll free its place.

Working offline

LaserCheck checks your licence online about once a month, in the background. If it can't get online, it carries on working for a further 14 days, then asks you to connect and click **Check now**. Nothing else needs the internet.

> LOST YOUR KEY?
>
> It's in your receipt email from Lemon Squeezy. You can also find it at **app.lemonsqueezy.com/my-orders** by signing in with your email.

15 Your data and privacy

Your documents stay on your computer. LaserCheck has no account, no cloud and no tracking.

When LaserCheck uses the internet

- **Installing**: the installer downloads the app's runtime once, from the Electron project, and checks its checksum.
- **Ask the document**: once, the AI pack is downloaded from the llama.cpp project (GitHub) and Hugging Face. Questions and answers stay on this computer.
- **Presenting with your phone**: only while you present and only if you choose Phone remote, LaserCheck serves a small remote-control page on your own Wi-Fi, locked with a random key. Nothing from the document is sent to the phone.

- **Translating**: a language pack is downloaded once from Mozilla for each pair of languages you use. Nothing from your documents is sent.
- **Sealing a document**: a fingerprint of the document (not the document itself) is sent to a timestamp service to get a trusted time.
- **Updates**: when LaserCheck starts and every few hours, a small file naming the newest version is fetched from the LaserCheck website, and the update itself when there is one. Nothing about you or your documents is sent. You can turn this off.
- **Your licence**: when you activate, and about once a month after, your licence key and this computer's name are sent to Lemon Squeezy to check the licence. Your documents never are.

That's all. Text recognition, the report checker and everything else run on your computer.

What's kept on this computer

- Your name (for comments and signatures) and, if you chose to remember it, your signature.
- Your details for **Fill in my details**.
- Pack templates, and the options you last used in dialogs.
- The report checker's settings, procedure limits and equipment register, and its history: photo fingerprints, small thumbnails and test equipment details from reports you've checked. Clear it from the checker's settings.
- Recent files and your signing certificate.
- Error reports: when something goes wrong, a note of the version, your system and where in LaserCheck's code it happened. Quoted text, numbers and document names are taken out, so it never holds your documents' contents. It isn't sent anywhere: if you tick **Offer me the report after a problem** (on the welcome card or **Help > Error Reports…**), LaserCheck offers it to copy or save for LaserCheck support after a problem. Clear it from the same place.

16 Troubleshooting

Activation says the key is in use on its maximum number of computers.

Move the licence off a computer you no longer use (see Moving to another computer), or get in touch.

Activation says the key was bought with a different email address.

Use exactly the email address on your receipt. It doesn't matter whether you use capital letters.

My Mac says the app or installer "can't be opened".

LaserCheck isn't signed with an Apple developer certificate. Use **System Settings → Privacy & Security → Open Anyway**, as in Installing.

Windows shows “Windows protected your PC”.

Click **More info**, then **Run anyway**.

I can't select or search the text in a PDF.

It's probably a scan. Use Recognise text.

Edited text looks slightly different from the original.

LaserCheck matches the original font as closely as it can. If the exact font isn't available, it uses the nearest match from its library. Pick a different font from the bar if you prefer.

Find & replace missed something.

The phrase may wrap onto a second line, or the page may be a scan.

Acrobat says my seal's signer is “not trusted”.

That's expected for a self-made certificate: see About certificates. The seal still proves the document hasn't changed.

A photo was rejected by the photo report.

It's probably HEIC (iPhone). Export it as JPEG.

The file got much bigger after saving.

Embedded fonts and pictures add size. Use **Reduce file size**, or **Save a Smaller Copy**.

17 About

LaserCheck version 1.34.0.

LaserCheck is built with open-source software, including Electron, PDF.js, pdf-lib, Tesseract.js, Human and TensorFlow.js (people detection), qrcode-generator and jsQR. The spelling dictionary is from SCOWL (word lists by Kevin Atkinson). The bundled fonts are licensed under the SIL Open Font License or Apache License; in the desktop app, **Help → Font Licences** shows the licence files.